

Math 140H — Honors Calculus I: Lecture Notes

Jan Reimann

2026-10-05

Table of contents

Welcome!	3
I Real Numbers	4
1 The Rational Numbers and the Number Line	5
1.1 From Counting to Measuring	5
1.2 The Number Line	6
1.3 Distances and Intervals	7
1.4 Incommensurable Lengths	8
1.5 Summary	9
2 The Real Numbers via Nested Intervals	10
2.1 Nested Intervals	10
2.2 The Nested Interval Axiom	11
2.3 Decimal Expansions	12
2.4 Other Bases	13
2.5 Summary	14
3 Jupyter Activity: p-ary Expansions	15
3.1 Before we start: how to use a Jupyter notebook	15
3.2 Importing Python modules	16
3.3 From a rational number to its p -ary expansion	17
3.4 Checking the examples	20
3.5 A few more instructive examples	20
3.6 Try it yourself	21
3.7 Food for thought	22
II Functions	23
4 The Concept of Function	24
4.1 Functions, Domain, and Range	24
4.2 Injective Functions	25
4.3 Monotone Functions	25
4.4 Even and Odd Functions	26

4.5	Building New Functions from Old	26
4.6	Summary	28
5	The Elementary Functions	29
5.1	Polynomials	29
5.2	Rational Functions	30
5.3	Algebraic and Power Functions	30
5.4	Exponential Functions	31
5.5	Logarithms	31
5.6	Trigonometric Functions	32
5.7	Summary	32
6	Continuity	34
6.1	The ε - δ Definition	35
6.2	Types of Discontinuities	36
6.3	Summary	37
7	Activity: Properties of Functions	38
7.1	Background	38
7.2	Step 1: Draw the graphs	38
7.3	Step 2: Make an educated guess	38
7.4	Verification: injectivity	39
7.5	Verification: even / odd	40
7.6	Verification: monotonicity	40
7.7	Continuity: the ε - δ game	40
7.8	Discontinuities	40
III	Limits	41
8	Sequences and Convergence	42
8.1	Sequences	42
8.2	The Limit of a Sequence	42
8.3	Convergence to a General Limit	43
8.4	Limit Laws	45
8.5	Comparison and the Squeeze Theorem	45
8.6	Monotone and Bounded Sequences	46
8.7	Applications	47
8.8	Summary	49
9	Convergence of Sequences: Reading Data	50
9.1	A Simple Query Function	53
9.2	Task 1: Explore the Data	53
9.3	Task 2: The ε Challenge	55

9.4	Dataset 2: Damped Oscillator	56
9.5	Task 3: Explore the Sequence	57
9.6	Task 4: The ε Challenge, Again	59
9.7	Task 5: Convergence of $\sqrt[n]{p}$	60
10	Limits of Functions	62
10.1	From Sequences to Functions	62
10.2	The ε - δ Version	63
10.3	Examples	64
10.4	One-Sided Limits	65
10.5	Limit Laws for Functions	66
10.6	Continuity of Combinations and Compositions	67
10.7	The Intermediate Value Theorem	68
10.8	Summary	68
IV	The Riemann Integral	70
11	Riemann Sums	71
11.1	Area of the Circle	71
11.2	Approximation of Areas by Rectangles	72
11.3	Equipartitions and the Definition of the Integral	73
11.4	Why the Choice Does Not Matter (optional)	74
11.5	Verifying the Riemann Integral	77
11.6	“New” Functions: x^2	78
11.7	Summary	78
12	Lab: Riemann Sums	80
12.1	Recap: the setup from lecture	85
12.2	Part 1: Start	86
12.3	Part 2: Left, right, and midpoint sums	88
12.4	Part 3: Refining the partition	91
12.5	Part 4: Back to the circle	93
12.6	Part 5: Experiment on your own	95
12.7	Part 6: when Riemann sums fail	96
13	Properties of the Integral	98
13.1	The Integral as Signed Area	98
13.2	Additivity	99
13.3	Integrals with $a \geq b$	100
13.4	Sums and Constant Multiples	101
13.5	Basic Rules for Integrals	102
13.6	Estimating Integrals	103

13.7 The Integral as a Mean Value	104
13.8 Summary	107
14 The Indefinite Integral	108
14.1 Definition and Basic Formula	108
14.2 Properties of Φ as a Function	109

Welcome!

These notes accompany Math 140H, Honors Calculus I, Fall 2026. Each lecture here corresponds to one class session and covers more or less the material presented in class, in the same order, so that it can serve as a compact study reference. HTML “hardcopies” of the Jupyter notebook activities from class are also included. Lecture notes will generally be available immediately **after** the corresponding class session.

State College, Fall 2026

Part I

Real Numbers

1 The Rational Numbers and the Number Line

Companion reading

This lecture corresponds to C&J §1.1a, together with the neighborhood definition from §1.1d — covered a bit early here since we will use it constantly for the rest of the semester.

Calculus is built on the real numbers, so before we can talk about limits, derivatives, or integrals we need to understand exactly what a real number *is* — and, just as importantly, why the rational numbers alone are not enough. This lecture makes that precise: we recall how the natural numbers extend to the rationals, set up the number line, and then prove that a length as ordinary as the hypotenuse of a simple right triangle cannot be a rational number at all.

1.1 From Counting to Measuring

There are two major uses for numbers: **counting** and **measuring**. The natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ are the tool for counting — abstract symbols for how many elements are in a collection, stripped of any reference to what is actually being counted. They come equipped with addition and multiplication satisfying a short list of familiar rules.

Remark 1.1. For all natural numbers a, b, c :

- $a + b = b + a$ and $ab = ba$ (commutative laws)
- $a + (b + c) = (a + b) + c$ and $a(bc) = (ab)c$ (associative laws)
- $a(b + c) = ab + ac$ (distributive law)
- $a + c = b + c \Rightarrow a = b$ (cancellation law)

These are the *rules* of arithmetic, not its *content* — everything else in this course is built on top of them, and we will not dwell on them further.

Natural numbers can be added and multiplied, but not always subtracted or divided. There is no natural number a with

$$5 + a = 2,$$

so subtraction is not always possible in $\mathbb{N}$. Extending $\mathbb{N}$ to the **integers** $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ fixes this: now $5 + (-3) = 2$.

In $\mathbb{Z}$, we still cannot always divide: there is no integer $b \in \mathbb{Z}$ with

$$5 \cdot b = 2.$$

Extending $\mathbb{Z}$ to the **rational numbers** fixes this in turn.

Definition 1.1.

A **rational number** is a number that can be written as p/q where $p, q \in \mathbb{Z}$ and $q \neq 0$. Every rational number has a unique representation p/q with $q > 0$ and $\gcd(p, q) = 1$ (no common factor larger than 1); we call this representation **in lowest terms**.

Within $\mathbb{Q}$, all four rational operations — addition, subtraction, multiplication, and division (except by 0) — can be carried out freely, and they obey exactly the same laws as in Remark 1.1. In this precise sense, $\mathbb{Q}$ is the smallest extension of the natural numbers in which counting-type arithmetic becomes measuring-type arithmetic.

1.2 The Number Line

To turn numbers into geometry, fix a line L , a point O on it called the **origin**, and a second point I to the right of O ; the segment OI serves as our unit scale for measuring — a “prototype meter.” The direction from O to I is called **positive**.

Every point on L is now uniquely determined by two pieces of information: its **distance** from O , and its **direction** (left or right) from O . For example, the point at distance $\frac{1}{2}$ in the positive direction represents the rational number $\frac{1}{2}$; the point at distance 2 in the positive direction represents 2.

Definition 1.2.

The distance of a number x from the origin is written $|x|$ and called the **absolute value** of x :

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0. \end{cases}$$

Note $|x| \geq 0$ always, with equality only when $x = 0$.

Every rational point can in fact be *constructed*, not just described: to locate p/q on L with ruler and compass, construct a subdivision of the segment OI into q equal parts, then extend the part closest to O , p times. We will identify each rational number with its representation on L .

1.3 Distances and Intervals

For $x, y \in L$, the number $|x - y|$ is the **distance between x and y** . Observe: if $x, y \in \mathbb{Q}$, then $|x - y| \in \mathbb{Q}$ as well — distances between rational points are themselves rational.

Definition 1.3.

For $a < b$, the **interval** between a and b comes in four flavors:

- open: $(a, b) := \{x : a < x < b\}$ (both endpoints excluded)
- closed: $[a, b] := \{x : a \leq x \leq b\}$ (both endpoints included)
- half-open: $[a, b) := \{x : a \leq x < b\}$ and $(a, b] := \{x : a < x \leq b\}$

Definition 1.4.

Given a number a and $\varepsilon > 0$, the **ε -neighborhood of a** is the open interval

$$\{y : |y - a| < \varepsilon\} = (a - \varepsilon, a + \varepsilon),$$

i.e. the set of points less than ε away from a .

We will meet ε -neighborhoods constantly once we get to limits — keep this definition close by.

1.3.1 Density

Proposition 1.1.

Between any two distinct rational numbers $a < b$ there is another rational number c with $a < c < b$. Consequently, between a and b there are infinitely many rational numbers.

Proof. Let $c = \frac{a+b}{2}$. Since $a, b \in \mathbb{Q}$ and $\mathbb{Q}$ is closed under addition and division by 2, $c \in \mathbb{Q}$; and since $a < b$, averaging gives $a < c < b$.

Now repeat the argument on the pair a, c to get a rational number strictly between a and c , hence strictly between a and b . Doing this again and again produces infinitely many rational numbers between a and b . $\square$

We say the rationals are **dense** on the number line: no matter how closely you look, you will always find another one.

1.4 Incommensurable Lengths

Given how constructible and how dense $\mathbb{Q}$ is, a natural question arises: is there anything “left in between” the rational numbers? Are there any *gaps*? Clearly, such a gap could not itself be an interval — density rules that out, since any interval of positive length contains infinitely many rational points. If a gap exists at all, it can only be a single missing point.

Remark 1.2. The discovery that not every length is a rational multiple of a chosen unit is due to the school of Pythagoras, in the fifth or sixth century BCE. Consider a right triangle with both legs of length 1; by the Pythagorean theorem, its hypotenuse a satisfies

$$1^2 + 1^2 = a^2 = 2,$$

and a can certainly be constructed with ruler and compass. The theorem below shows this length cannot be written as p/q for integers p, q — the hypotenuse and the legs of this triangle are, in the language of the ancients, **incommensurable**: there is no common unit of which both are integer multiples.

Theorem 1.1.

There is no rational number x with $x^2 = 2$. Equivalently, $\sqrt{2} \notin \mathbb{Q}$.

Proof. Suppose, toward a contradiction, that $\sqrt{2} = p/q$ for integers p, q with $q \neq 0$ and $\gcd(p, q) = 1$ (i.e. p/q is in lowest terms).

1. Square both sides and clear the denominator: $p^2 = 2q^2$.
2. Conclude that p^2 is even — if p were odd, p^2 would be odd, but $p^2 = 2q^2$ is even. Write $p = 2r$ for some integer r .
3. Substitute $p = 2r$ into the equation from Step 1 and simplify to get an equation for q^2 in terms of r : $(2r)^2 = 2q^2 \Rightarrow 4r^2 = 2q^2 \Rightarrow q^2 = 2r^2$.
4. Conclude that q is also even, by the same reasoning as in Step 2.
5. Steps 2 and 4 show that 2 divides both p and q . This contradicts $\gcd(p, q) = 1$, which we assumed when we wrote p/q in lowest terms.
6. Conclude: $\sqrt{2}$ is irrational — no integers p, q satisfy $\sqrt{2} = p/q$, so $\sqrt{2} \notin \mathbb{Q}$.

□

Exercise 1.1. Adapt the proof of Theorem 1.1 to show that $\sqrt{3}$ is irrational. Then try to see where the argument would break if you attempted the same proof for $\sqrt{4}$ — since $\sqrt{4} = 2$ is certainly rational, something in the argument *must* fail. Where, exactly?

1.5 Summary

- The rationals $\mathbb{Q}$ satisfy all the usual laws of arithmetic (Remark 1.1) and are **dense**: between any two rationals lies another (Proposition 1.1).
- Distances $|x - y|$, intervals, and ε -neighborhoods (Definition 1.3, Definition 1.4) give us the basic vocabulary for talking about “closeness” on the number line — vocabulary we will lean on heavily once limits arrive.
- Density notwithstanding, $\mathbb{Q}$ does not exhaust the number line: Theorem 1.1 exhibits an ordinary, constructible length that is not a rational number.
- The rationals therefore leave *gaps* in the line. Lecture 2 closes these gaps rigorously, using **nested intervals** to construct the real numbers.

2 The Real Numbers via Nested Intervals

Companion reading

This lecture corresponds to C&J §1.1b–c.

Last time we found a gap: the length $\sqrt{2}$ is constructible with ruler and compass, yet Theorem 1.1 shows it is not a rational number. So what *is* $\sqrt{2}$, as a number? We are used to describing such quantities through infinite decimal expansions, e.g.

$$\pi = 3.1415926 \dots,$$

but so far this is just a string of digits — it is not clear what it *means* mathematically. The key idea of this lecture is that an infinite decimal (or any other description of an irrational number) is really a recipe for **approximating** a point by rational numbers, to any desired degree of accuracy. We make this precise using shrinking intervals with rational endpoints.

2.1 Nested Intervals

Suppose x is a point on the number line that we want to pin down using only rational numbers. Start with any closed interval $I_1 = [a_1, b_1]$ with $a_1, b_1 \in \mathbb{Q}$ and $a_1 \leq x \leq b_1$. Now choose a subinterval $I_2 = [a_2, b_2] \subseteq I_1$ that still contains x :

$$a_1 \leq a_2 \leq x \leq b_2 \leq b_1,$$

for instance by splitting I_1 in half and keeping whichever half contains x . Repeating this forever produces a chain

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots, \quad x \in I_n \text{ for every } n.$$

For this to genuinely *pin down* x — rather than just hover near it — we need one more ingredient: the intervals must shrink all the way down to a point.

Definition 2.1.

A **nested sequence of closed intervals with rational endpoints** is a sequence $I_n = [a_n, b_n]$, $n = 1, 2, 3, \dots$, with $a_n, b_n \in \mathbb{Q}$, such that

1. $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$, and
2. the lengths $|I_n| := b_n - a_n$ tend to 0 as $n \rightarrow \infty$: for every $\varepsilon > 0$ there is an N such that $|I_n| < \varepsilon$ for all $n \geq N$.

Proposition 2.1.

If a nested sequence of closed intervals (I_n) contains a point in every I_n , that point is unique: there is at most one x with $x \in I_n$ for all n .

Proof. Suppose x and y both lie in every I_n , and toward a contradiction suppose $x \neq y$. Let $\varepsilon_0 = |x - y| > 0$. Since $|I_n| \rightarrow 0$, there is some n with $|I_n| < \varepsilon_0$. But $x, y \in I_n$ means $|x - y| \leq |I_n| < \varepsilon_0 = |x - y|$, which is absurd. Hence $x = y$. $\square$

So a nested sequence of intervals can “address” at most one point — and, geometrically, every point x on the line does arise this way:

Theorem 2.1.

Every real number x is the unique point contained in every interval of some nested sequence of closed intervals with rational endpoints.

Proof. Uniqueness is Proposition 2.1. For existence, pick any rational bracket $I_1 = [a_1, b_1]$ with $a_1 \leq x \leq b_1$, and repeatedly bisect: given $I_n = [a_n, b_n]$, let $m_n = \frac{a_n + b_n}{2}$ (rational, since $a_n, b_n \in \mathbb{Q}$), and set

$$I_{n+1} = \begin{cases} [a_n, m_n] & \text{if } x \leq m_n \\ [m_n, b_n] & \text{if } x > m_n. \end{cases}$$

Then $x \in I_{n+1} \subseteq I_n$ and $|I_{n+1}| = |I_n|/2$, so $|I_n| = |I_1|/2^{n-1} \rightarrow 0$. This is a nested sequence of closed intervals with rational endpoints containing x . $\square$

In this sense, every real number can be described — completely and unambiguously — by an infinite sequence of rational numbers (the endpoints a_n, b_n).

2.2 The Nested Interval Axiom

The construction above starts from a real number x and produces a nested sequence around it. What about the converse: if we are simply *handed* a nested sequence of closed intervals with rational endpoints, must there be a point in $\bigcap_n I_n$ at all?

For **open** intervals, the answer is no: $I_n = (0, \frac{1}{n})$ satisfies $I_1 \supseteq I_2 \supseteq \dots$ and $|I_n| = \frac{1}{n} \rightarrow 0$, yet no point lies in every I_n — any candidate $x > 0$ fails once $n > 1/x$, and $x = 0$ is excluded from every I_n by openness.

For **closed** intervals, this is far less obvious, and cannot be deduced from the algebraic properties of $\mathbb{Q}$ alone (after all, $\mathbb{Q}$ itself has a gap at $\sqrt{2}$). We take it as a basic postulate about the number line.

Axiom (Nested Interval Property)

If $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$ is a nested sequence of closed intervals with rational endpoints, then

$$\bigcap_{n \in \mathbb{N}} I_n \neq \emptyset,$$

i.e. there is a point x contained in every I_n .

This is a genuine **axiom**: a basic postulate we accept without proof, rather than a consequence of things we already know. Together with Proposition 2.1, it guarantees that no gaps exist on the number line — every nested sequence of rational brackets zeroes in on exactly one real number, no more and no less. This completeness of the number line is fundamental for everything that follows in this course; we will meet it again, in other guises, once we get to limits.

2.3 Decimal Expansions

Decimal expansions are simply a very concrete, systematic way of producing a nested sequence of intervals — one that subdivides by tens at every step.

Definition 2.2.

Let x be a point on the number line. Set

$$c_0 := \lfloor x \rfloor,$$

the greatest integer with $c_0 \leq x \leq c_0 + 1$, and $I_0 = [c_0, c_0 + 1]$, so $|I_0| = 1$.

Divide I_0 into ten equal parts using the points $c_0 + \frac{1}{10}, c_0 + \frac{2}{10}, \dots, c_0 + \frac{9}{10}$, and let $c_1 \in \{0, 1, \dots, 9\}$ be the digit with

$$c_0 + \frac{c_1}{10} \leq x \leq c_0 + \frac{c_1}{10} + \frac{1}{10},$$

and set I_1 to be this interval, so $|I_1| = \frac{1}{10}$.

Iterating, having chosen digits $c_1, \dots, c_{n-1} \in \{0, \dots, 9\}$, subdivide I_{n-1} into ten equal parts and let $c_n \in \{0, \dots, 9\}$ be the digit with

$$c_0 + \frac{c_1}{10} + \dots + \frac{c_n}{10^n} \leq x \leq c_0 + \frac{c_1}{10} + \dots + \frac{c_n}{10^n} + \frac{1}{10^n},$$

and let I_n be this interval, so $|I_n| = 10^{-n}$.

By construction $(I_n)_{n \geq 0}$ is exactly a nested sequence of closed intervals with rational endpoints in the sense of Definition 2.1: $I_0 \supseteq I_1 \supseteq I_2 \supseteq \dots$, each contains x , and $|I_n| = 10^{-n} \rightarrow 0$. We abbreviate this whole sequence by writing

$$x = c_0 + 0.c_1c_2c_3 \dots,$$

the **decimal expansion** of x — a notational shortcut for the nested intervals, not a new kind of number. (Careful when $c_0 < 0$: the digits after the point are still *added* to c_0 . For instance $x = -2.75$ has $c_0 = \lfloor -2.75 \rfloor = -3$ and $x = -3 + 0.25$, not $-3 - 0.25$.)

Combining Definition 2.2 with Proposition 2.1 and the Nested Interval Axiom:

- Every real number has an (infinite) decimal expansion.
- Conversely, every such expansion — i.e. every choice of $c_0 \in \mathbb{Z}$ and digits $c_1, c_2, \dots \in \{0, \dots, 9\}$ — is a nested sequence of closed intervals with rational endpoints, so by the axiom it defines a real number.

Decimal expansions are not always unique

Some real numbers have *two* decimal expansions: for instance

$$1.000 \dots = 0.999 \dots$$

Exercise 2.1. Why does $1.000 \dots = 0.999 \dots$? (Hint: what would $1 - 0.999 \dots$ have to be, if it were positive?) More generally, can you characterize *exactly* which real numbers have two decimal expansions?

Solution. If $1 - 0.999 \dots = \delta > 0$, then δ would have to be smaller than 10^{-n} for every n (since $0.\underbrace{9 \dots 9}_n = 1 - 10^{-n}$ already exceeds $1 - \delta$), forcing $\delta = 0$ — contradiction. So the two expansions name the same point.

In general, a real number has two decimal expansions exactly when it is a **terminating** decimal, i.e. of the form $m/10^n$ for some $m \in \mathbb{Z}$, $n \geq 0$: one expansion ends in all 0's, the other in all 9's. Every other real number has a unique decimal expansion.

2.4 Other Bases

Nothing in Definition 2.2 is special about the number 10 — we could subdivide each interval into any number $p > 1$ of equal parts instead of ten, producing digits $c_i \in \{0, 1, \dots, p-1\}$ and a **base- p (or p -ary) expansion** of x .

The case $p = 2$, the **binary expansion**, is exactly the bisection procedure from the proof of Theorem 2.1: at each step we only need to decide whether x lies in the lower or upper half of the current interval.

Example 2.1.

Let's find the first three binary digits of $x = 0.76$. Start with $I_0 = [0, 1]$, so $c_0 = 0$.

- Midpoint 0.5: since $0.76 \geq 0.5$, digit $c_1 = 1$ and $I_1 = [0.5, 1]$.
- Midpoint 0.75: since $0.76 \geq 0.75$, digit $c_2 = 1$ and $I_2 = [0.75, 1]$.
- Midpoint 0.875: since $0.76 < 0.875$, digit $c_3 = 0$ and $I_3 = [0.75, 0.875]$.

So $0.76 = 0.110\dots$ in base 2.

Exercise 2.2. Find the first three fractional digits of the base-2 expansion of $x = 0.3$, and the first three fractional digits of the base-3 expansion of $x = 0.76$.

2.5 Summary

- A **nested sequence of closed intervals with rational endpoints** (Definition 2.1) can contain at most one point (Proposition 2.1), and every real number arises as the unique point of some such sequence (Theorem 2.1).
- The converse — that every nested sequence of closed intervals with rational endpoints *does* contain a point — is not provable from what came before; it is the **Nested Interval Axiom**, the basic postulate that guarantees there are no gaps in the number line.
- **Decimal expansions** (Definition 2.2) are the special case of base-10 subdivision; every real number has one, though numbers of the form $m/10^n$ have exactly two (Exercise 2.1).
- The same construction works in any base $p > 1$, giving p -ary expansions; base 2 is exactly the bisection algorithm behind Theorem 2.1.

3 Jupyter Activity: p-ary Expansions

Math 140H, Honors Calculus I

3.1 Before we start: how to use a Jupyter notebook

This is the first notebook of the semester, so we start a quick orientation:

- A notebook is a sequence of **cells**. This text lives in a *markdown cell* (formatted text); cells with **code** in the toolbar are *code cells* (Python).
- Click a cell and press **Shift+Enter** to run it and move to the next one. Code cells must be run in order the first time through, top to bottom — a later cell may use a function or variable that an earlier cell defined.
- If something breaks, it is often because a cell was skipped or run twice out of order. Use the menu command *Run All* (or *Restart Kernel and Run All*) to run the whole notebook cleanly from the top.
- You do **not** need to know Python already. Every function below is commented line by line — read the comments (the lines starting with **#**) as a description of the math, and the code as one particular way of writing it down.

3.1.1 Code cells

You can think of a code cell very much like a fancy calculator. Move to the next cell and run it.

```
16+(53*8.7777-128)
```

353.2181

Careful: In Python, exponentiation is ******, not **^**.

```
2**4
```

16

```
2^4
```

6

`^` in Python is the operation called **bitwise XOR**; that explains the strange result above.

We will learn more about Jupyter notebooks (and a little bit more about Python) in the course of this semester. Today, we will simply experiment a bit with pre-defined Python function that converts rational numbers to p -ary expansions. Go ahead and run the cells below one after the other (do not forget to read the text inbetween.)

3.2 Importing Python modules

A lot functionality in Python is provided by *modules*. The following cell loads the modules we need for this notebook.

```
# We import three tools before doing anything else:
#
#   math      -- the standard library module with math.floor(y), the greatest integer <=
#               (We need floor of *exact fractions*, not of floating-point approximation
#               see the note below.)
#   Fraction  -- from the 'fractions' module. A Fraction stores a rational number as an e
#               numerator/denominator pair, e.g. Fraction(1, 3) is *exactly* one third,
#               This matters a lot here: a computer's ordinary decimal numbers (called '
#               are themselves binary approximations, so 1/3 as a float is already sligh
#               and the tiny error would snowball after enough steps of the algorithm be
#               Using Fraction means every digit we compute is provably correct, not jus
#   numpy     -- imported as 'np' by convention. We use it only to store the list of digi
#               compute as a numpy array, which is a convenient container for a sequence
import math
from fractions import Fraction
import numpy as np
```

The cells below show the advantage of the `Fraction` function:

```
1/2+1/3 # will use decimal expansions
```

```
0.8333333333333333
```

```
Fraction(1,2)+Fraction(1,3) # will preserve the precise fraction representation
```

```
Fraction(5, 6)
```

3.3 From a rational number to its p -ary expansion

We generalize the nested interval construction to base p : given x , set

$$c_0 := \lfloor x \rfloor,$$

and then repeatedly zoom in. Having peeled off $c_0, c_1, \dots, c_{n-1}$, look at what is left over,

$$r_{n-1} = x - c_0 - \frac{1}{p}c_1 - \dots - \frac{1}{p^{n-1}}c_{n-1} \in [0, p^{-(n-1)}),$$

multiply it by p^{n-1} so it lands back in $[0, 1)$, and read off the next digit as the integer part of p times that:

$$c_n = \lfloor p \cdot (p^{n-1}r_{n-1}) \rfloor.$$

(Multiplying by p here means zooming in by a factor of p .)

The function below does exactly this, but it tracks the *rescaled* remainder (called **remainder** in the code, always kept in $[0, 1)$) rather than r_{n-1} itself, since that is what you actually multiply by p at each step:

1. **remainder** = **x** - **floor(x)**, the fractional part of x , in $[0, 1)$.
2. Repeat: multiply **remainder** by p . Its integer part is the next digit; subtract that integer part off, leaving a new **remainder** back in $[0, 1)$, and go again.

```

def rational_to_p_ary(x, p, max_digits=50):
    """
    Compute the base-p expansion of a rational number x.

    Parameters
    -----
    x : Fraction, int, or a string like "7/12"
        The rational number to expand. (Do NOT pass a plain float such as 0.76 -- floats
        already binary approximations, so we would be expanding the wrong number. Use
        Fraction(76, 100) instead. See the discussion above.)
    p : int
        The base, an integer > 1 (p = 2 for binary, p = 10 for the usual decimal expansion)
    max_digits : int, optional (default = 50)
        how many fractional digits to compute

    Returns
    -----
    integer_part : int
        c_0, the integer part of x (i.e. floor(x)).
    digits : numpy array of int
        The fractional digits c_1, c_2, c_3, ... that were computed, in order.
    repeat_start : int or None
        If the expansion is eventually periodic, the index (0-based, into `digits`) where
        repeating block begins. If the expansion terminates exactly (all later digits are 0)
        this is None.
    """
    if not isinstance(p, int) or p <= 1:
        raise ValueError("the base p must be an integer greater than 1")

    # Convert x to an exact fraction, no matter what form it was given in.
    x = Fraction(x)

    # Step 1: peel off the integer part c_0 = floor(x). What remains, x - c_0, is the
    # 'fractional part' of x and always lies in [0, 1).
    integer_part = math.floor(x)
    remainder = x - integer_part

    digits = []          # the digits c_1, c_2, ... found so far, in order
    seen_remainders = {} # remembers, for each remainder value we've seen, which step
                        # first showed up at -- this is how we notice a repeat

    repeat_start = None

```

```

for step in range(max_digits):
    # Zooming into the next sub-interval of
    # length 1/p means multiplying the remainder by p; the integer part of the result
    # exactly which of the p equal pieces x fell into, i.e. the next digit.
    remainder = remainder * p
    next_digit = math.floor(remainder)
    digits.append(next_digit)

    # Subtract off the digit we just read, leaving a fresh remainder back in [0, 1),
    # ready for the next iteration.
    remainder = remainder - next_digit

return integer_part, np.array(digits, dtype=int), repeat_start

```

The cell below defines a small helper function to print an expansion in a readable form. We deliberately write it as $c_0 + 0.c_1c_2c_3\dots$ with an explicit $+$, matching the lecture notes' warning: if c_0 is negative, the fractional digits are still *added* to it, not subtracted, so -3.25 written the ordinary way would be dangerously ambiguous. Writing $-3 + 0.25$ instead leaves no room for that mistake.

```

def format_expansion(integer_part, digits, p, repeat_start=None):
    """
    Turn the output of rational_to_p_ary into a readable string, e.g.
        format_expansion(0, np.array([1, 1, 0]), 2, None) -> '0 + 0.110 (base 2)'
        format_expansion(0, np.array([3]), 10, 0)          -> '0 + 0.(3) (base 10)' #
    """
    # Turn each digit into a one-character string. In any base p <= 10 these are ordinary
    # for p > 10 we would need letters (like hexadecimal's a-f), which we don't handle here
    digit_chars = [str(int(d)) for d in digits]

    if repeat_start is None:
        # No repeating block: just list every digit we have.
        fractional_part = "".join(digit_chars) if len(digit_chars) > 0 else "0"
    else:
        # Split into the digits before the repeat (head) and the repeating block itself
        # and wrap the tail in parentheses to mark it as repeating forever.
        head = "".join(digit_chars[:repeat_start])
        tail = "".join(digit_chars[repeat_start:])
        fractional_part = f"{head}({tail})" if head else f"({tail})"

    return f"{integer_part} + 0.{fractional_part} (base {p})"

```

3.4 Checking the examples

In the lecture notes, we found the first three *binary* digits of $x = 0.76$ by hand: $c_1 = 1$, $c_2 = 1$, $c_3 = 0$, i.e. $0.76 = 0.110\dots$ in base 2. Let's confirm that.

```
integer_part, digits, repeat_start = rational_to_p_ary(Fraction(76, 100), 2, max_digits=20)
print(format_expansion(integer_part, digits, 2, repeat_start))
print("first three fractional digits:", digits[:3], " -- matches c1, c2, c3 = 1, 1, 0 from Example 2.6")
```

```
0 + 0.110000101000111101011100001010 (base 2)
first three fractional digits: [1 1 0] -- matches c1, c2, c3 = 1, 1, 0 from Example 2.6
```

Now find the first three fractional digits of the base-2 expansion of $x = 0.3$, and of the base-3 expansion of $x = 0.76$. Work those out on paper first – then run the cell below to check yourself.

```
ip, d, r = rational_to_p_ary(Fraction(3, 10), 2, max_digits=20)
print("0.3 in base 2:", format_expansion(ip, d, 2, r))
print("first three digits:", d[:3])
print()
ip, d, r = rational_to_p_ary(Fraction(76, 100), 3, max_digits=20)
print("0.76 in base 3:", format_expansion(ip, d, 3, r))
print("first three digits:", d[:3])
```

```
0.3 in base 2: 0 + 0.01001100110011001100 (base 2)
first three digits: [0 1 0]
```

```
0.76 in base 3: 0 + 0.20211200100201102212 (base 3)
first three digits: [2 0 2]
```

3.5 A few more instructive examples

- $1/3$ in base 10 has *no* terminating part at all – the repeating block starts immediately.
- $1/6$ in base 10 has one non-repeating digit followed by a repeating block (matches $1/6 = 0.1\bar{6}$).
- $1/10$ terminates in base 10 (as expected, $1/10 = 0.1$ exactly) but is *purely repeating* in base 2 – this is the mathematical reason 0.1 cannot be stored exactly as an ordinary floating-point number, and is why we insisted on `Fraction` above instead of Python floats.

- Negative x : $x = -11/4 = -2.75$. Its integer part is $c_0 = \lfloor -2.75 \rfloor = -3$, and the digits describe 0.25, which gets *added* to -3 (giving back -2.75), not subtracted.

```
for label, value, base, n in [
    ("1/3", Fraction(1, 3), 10, 10),
    ("1/6", Fraction(1, 6), 10, 10),
    ("1/10", Fraction(1, 10), 10, 10),
    ("1/10", Fraction(1, 10), 2, 30),
    ("-11/4", Fraction(-11, 4), 10, 10),
]:
    ip, d, r = rational_to_p_ary(value, base, max_digits=n)
    print(f"{label:>6} = {format_expansion(ip, d, base, r)}")
```

```
1/3 = 0 + 0.3333333333 (base 10)
1/6 = 0 + 0.1666666666 (base 10)
1/10 = 0 + 0.1000000000 (base 10)
1/10 = 0 + 0.000110011001100110011001100110 (base 2)
-11/4 = -3 + 0.2500000000 (base 10)
```

3.6 Try it yourself

Pick your own rational number x and a base $p > 1$. Use `rational_to_p_ary` to get its p -ary expansion.

```
# Replace the values below with your own choice of x (as a Fraction), p.
x = Fraction(1, 2)
p = 10

ip, d, r = rational_to_p_ary(x, p, max_digits=30)
print(f"x in base {p}:", format_expansion(ip, d, p, r))
```

```
x in base 10: 0 + 0.500000000000000000000000000000 (base 10)
```

3.7 Food for thought

Question 1: For a fixed base p , which fractions will have a p -ary expansion that will eventually be identically 0? (such as $1/2$ in base 10) Try to formulate a hypothesis and test it using the cell above.

Question 2: Is it true that all p -ary expansions of rational numbers eventually become periodic? If yes, why is this the case?

Part II

Functions

4 The Concept of Function

With the number line in hand, we can start doing calculus — but calculus is not really about individual numbers. It is about *dependence*: how one quantity changes as another one does. The area of a circle depends on its radius, the height of a falling stone depends on the elapsed time, the pressure of a gas depends on its volume. The mathematical object that captures such a dependence is a **function**, and getting comfortable with the vocabulary now will pay off for the rest of the course.

4.1 Functions, Domain, and Range

Definition 4.1.

A **function** f is a rule that assigns to every element x of a set D — the **domain** of f — one single, unambiguously determined object $f(x)$, its **value** at x , lying in a set E — the **codomain**. We write

$$f: D \rightarrow E, \quad x \mapsto f(x).$$

In this course the domain is almost always a subset of the number line, $D \subseteq \mathbb{R}$, and the codomain is $E = \mathbb{R}$; we then call f a **real-valued function of a real variable** and write simply $f: D \rightarrow \mathbb{R}$.

The word *unambiguously* is the heart of the definition: for each input $x \in D$ there must be **exactly one** output $f(x)$ — no more (that would not be a rule) and no fewer (every point of the domain must get a value). The domain is part of the data of a function: $x \mapsto x^2$ on $\mathbb{R}$ and $x \mapsto x^2$ on $[0, \infty)$ are, strictly speaking, different functions.

Remark 4.1. When a function is specified by a formula with no domain stated, we adopt the convention that its domain is the **natural domain**: the largest set of real numbers for which the formula makes sense. For $x \mapsto 1/(x-1)$ this is $\mathbb{R} \setminus \{1\}$; for $x \mapsto \sqrt{x}$ it is $[0, \infty)$. We are always free to *restrict* to a smaller domain when we want to.

Definition 4.2.

The **range** (or **image**) of $f: D \rightarrow \mathbb{R}$ is the set of values it actually attains:

$$\text{ran}(f) = \{y \in \mathbb{R} : y = f(x) \text{ for some } x \in D\}.$$

Remark 4.2. The range need not fill up the codomain. For $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, the codomain is $\mathbb{R}$ but

$$\text{ran}(f) = [0, \infty),$$

since squares are never negative and every $y \geq 0$ is $(\sqrt{y})^2$. Saying “ f maps into $\mathbb{R}$ ” is a statement about where the values are *allowed* to live; the range records where they *actually* land.

4.2 Injective Functions

Definition 4.3.

A function $f: D \rightarrow \mathbb{R}$ is **injective** (or **one-to-one**) if it sends distinct inputs to distinct outputs:

$$x \neq y \quad \Rightarrow \quad f(x) \neq f(y),$$

or equivalently, taking the contrapositive, $f(x) = f(y) \quad \Rightarrow \quad x = y$.

Example 4.1.

- $f(x) = x^3$ is injective on $\mathbb{R}$: if $x^3 = y^3$ then $x = y$ (taking cube roots).
- $g(x) = x^2$ is **not** injective on $\mathbb{R}$: $g(-1) = 1 = g(1)$, yet $-1 \neq 1$. Restricted to $[0, \infty)$, however, g is injective.

Injectivity is exactly the property that lets us run a function backwards; we return to this when we discuss inverse functions below.

4.3 Monotone Functions

Definition 4.4.

Let $f: D \rightarrow \mathbb{R}$ with $D \subseteq \mathbb{R}$. For all $x, w \in D$, we say f is

- **increasing** if $x \leq w \quad \Rightarrow \quad f(x) \leq f(w)$;
- **strictly increasing** if $x < w \quad \Rightarrow \quad f(x) < f(w)$;
- **decreasing** if $x \leq w \quad \Rightarrow \quad f(x) \geq f(w)$;
- **strictly decreasing** if $x < w \quad \Rightarrow \quad f(x) > f(w)$.

A function of any of these four types is called **monotone**.

Example 4.2.

- $f(x) = x^3$ is strictly increasing on all of $\mathbb{R}$.

- $g: [0, \pi] \rightarrow \mathbb{R}$, $g(x) = \cos x$, is strictly decreasing: as the angle runs from 0 to π , the x -coordinate on the unit circle drops steadily from 1 to -1 .

Exercise 4.1. Show that a strictly increasing function is injective (and likewise for strictly decreasing). Is the converse true — must an injective function be monotone?

4.4 Even and Odd Functions

Definition 4.5.

Suppose the domain D is symmetric about 0, i.e. $x \in D \Rightarrow -x \in D$. A function $f: D \rightarrow \mathbb{R}$ is

- **even** if $f(-x) = f(x)$ for all $x \in D$ — its graph is symmetric about the y -axis;
- **odd** if $f(-x) = -f(x)$ for all $x \in D$ — its graph is symmetric about the origin.

Example 4.3.

x^2 , $|x|$, and $\cos x$ are even; x , x^3 , and $\sin x$ are odd; $x + x^2$ is neither. Most functions are neither even nor odd.

Exercise 4.2. Show that every function f on a symmetric domain can be written *uniquely* as a sum $f = f_e + f_o$ of an even function f_e and an odd function f_o . (Hint: what must $\frac{1}{2}(f(x) + f(-x))$ and $\frac{1}{2}(f(x) - f(-x))$ be?)

4.5 Building New Functions from Old

Most functions we work with are assembled from simpler ones by a few standard operations.

4.5.1 Arithmetic combinations

Definition 4.6.

Let $f, g: D \rightarrow \mathbb{R}$ be two functions on the same domain, and $c \in \mathbb{R}$. Then

$$f + g, \quad cf, \quad f \cdot g$$

are again functions on D , defined **pointwise**: $(f + g)(x) = f(x) + g(x)$, $(cf)(x) = cf(x)$, $(f \cdot g)(x) = f(x)g(x)$. The **quotient** f/g , given by $(f/g)(x) = f(x)/g(x)$, is a function on the smaller domain

$$D \setminus \{x \in D : g(x) = 0\}.$$

4.5.2 Composition

Definition 4.7.

Let $g: D \rightarrow \mathbb{R}$ and $f: D' \rightarrow \mathbb{R}$ with $\text{ran}(g) \subseteq D'$. The **composition** $f \circ g$ (“first g , then f ”) is

$$(f \circ g)(x) := f(g(x)), \quad x \in D,$$

a function with domain $\text{dom}(f \circ g) = \text{dom}(g)$.

Example 4.4.

With $f(x) = \sqrt{x}$ and $g(x) = 1 + x^2$:

$$(f \circ g)(x) = \sqrt{1 + x^2} \quad (\text{all } x \in \mathbb{R}), \quad (g \circ f)(x) = 1 + x \quad (x \geq 0).$$

The order matters: $f \circ g$ and $g \circ f$ are different functions.

4.5.3 Inverse functions

Definition 4.8.

If $f: D \rightarrow \mathbb{R}$ is injective, then for each $y \in \text{ran}(f)$ there is **exactly one** $x \in D$ with $f(x) = y$. Sending each such y to that x defines the **inverse function**

$$f^{-1}: \text{ran}(f) \rightarrow D, \quad f^{-1}(f(x)) = x \text{ for all } x \in D, \quad f(f^{-1}(y)) = y \text{ for all } y \in \text{ran}(f).$$

Its domain is $\text{ran}(f)$ and its range is D .

Notation trap

f^{-1} denotes the **inverse function**, not the reciprocal $1/f$. For $f(x) = x^3$ we have $f^{-1}(y) = \sqrt[3]{y}$, whereas $\frac{1}{f(x)} = x^{-3}$.

Remark 4.3. The point (a, b) lies on the graph of f exactly when (b, a) lies on the graph of f^{-1} . So the graph of f^{-1} is the mirror image of the graph of f across the line $y = x$.

Example 4.5.

$f(x) = x^3$ is injective on $\mathbb{R}$ with $\text{ran}(f) = \mathbb{R}$, and $f^{-1}(y) = \sqrt[3]{y}$. By contrast $g(x) = x^2$ on $\mathbb{R}$ has no inverse (it is not injective); restricted to $[0, \infty)$ it does, namely $g^{-1}(y) = \sqrt{y}$.

4.6 Summary

- A **function** $f: D \rightarrow \mathbb{R}$ (Definition 4.1) assigns to each point of its **domain** D exactly one real value; the set of values attained is its **range** (Definition 4.2), which may be a proper subset of the codomain.
- f is **injective** (Definition 4.3) if distinct inputs give distinct outputs, and **monotone**
 - (1) if it consistently preserves or reverses order; strict monotonicity implies injectivity (Exercise 4.1).
- On a symmetric domain, f may be **even** or **odd** (Definition 4.5), i.e. symmetric about the y -axis or the origin.
- New functions are built from old ones by pointwise **arithmetic** (Definition 4.6), **composition** (Definition 4.7), and — when f is injective — by forming the **inverse** f^{-1} (Definition 4.8), whose graph is the reflection of that of f across $y = x$.

5 The Elementary Functions

Companion reading

This lecture corresponds to C&J §1.3.

Almost every function met in a first calculus course is built out of a short catalogue of standard functions: polynomials, rational functions, powers, exponentials, logarithms, and the trigonometric functions. These are the **elementary functions**. This lecture is a tour of the list, noting for each type its natural domain, its range, and its basic shape — information we will need constantly once we start differentiating and integrating.

5.1 Polynomials

Definition 5.1.

A **polynomial** is a function of the form

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n,$$

with real coefficients $a_0, a_1, \dots, a_n$. If $a_n \neq 0$, the integer n is the **degree** of p .

The lowest degrees have familiar names:

degree	form	name
0	$y = a_0$	constant
1	$y = a_0 + a_1x$	linear
2	$y = a_0 + a_1x + a_2x^2$	quadratic
3	$y = a_0 + a_1x + a_2x^2 + a_3x^3$	cubic

A polynomial is a finite sum of constant multiples of the **power functions** $1, x, x^2, x^3, \dots$, each of which is defined for every real number. Hence every polynomial has natural domain

$$\text{dom}(p) = \mathbb{R}$$

(unless we deliberately restrict it).

Remark 5.1. The pure powers x^n share a few features worth remembering. Every graph passes through $(0, 0)$ and $(1, 1)$. On the interval $[0, 1]$ a higher power lies *below* a lower one ($x^3 \leq x^2 \leq x$), while on $[1, \infty)$ the order reverses. The power x^n is even when n is even and odd when n is odd; for odd n it is strictly increasing on $\mathbb{R}$ (hence injective), while for even n it is not injective on $\mathbb{R}$.

5.2 Rational Functions

Definition 5.2.

A **rational function** is a quotient of two polynomials,

$$r(x) = \frac{p(x)}{q(x)}, \quad q \not\equiv 0.$$

Its natural domain is $\mathbb{R} \setminus Z$, where

$$Z = \{x \in \mathbb{R} : q(x) = 0\}$$

is the (finite) set of zeros of the denominator.

Example 5.1.

- $\frac{x}{x^2 + 1}$ is a rational function. Since $x^2 + 1 > 0$ for every real x , the set Z is empty and the domain is all of $\mathbb{R}$.
- $\frac{\sqrt{x}}{x^3 - x}$ is **not** a rational function: $\sqrt{x}$ is not a polynomial. (It is, however, still an elementary function, as it is the arithmetic combination of two elementary functions - see next section)

5.3 Algebraic and Power Functions

Definition 5.3.

A **power function** is $f(x) = x^\alpha$ with a fixed real exponent $\alpha \in \mathbb{R}$. For $x > 0$ this is defined for every α (via $x^\alpha = \exp(\alpha \ln x)$, once we have $\exp$ and $\ln$). For an **integer** exponent the domain extends to

$$\text{dom}(x^\alpha) = \begin{cases} \mathbb{R} & \alpha \geq 0, \\ \mathbb{R} \setminus \{0\} & \alpha < 0. \end{cases}$$

Remark 5.2. For a non-integer exponent one usually keeps the domain $[0, \infty)$ (or $(0, \infty)$ if $\alpha < 0$). The roots $x^{1/2} = \sqrt{x}$, $x^{1/3} = \sqrt[3]{x}$, ... are examples; odd roots such as $\sqrt[3]{x}$ can in fact be extended to all of $\mathbb{R}$. Power functions with rational exponent, and more generally functions obtained from polynomials by the four arithmetic operations together with root extraction, are called **algebraic functions**. The exponential and logarithm below are *not* algebraic — they are **transcendental**.

5.4 Exponential Functions

Definition 5.4.

Fix a **base** $a > 0$. The **exponential function** with base a ,

$$\exp_a : \mathbb{R} \rightarrow \mathbb{R}, \quad \exp_a(x) = a^x,$$

is defined for every real x . For $a \neq 1$ it takes only positive values, and every positive value is attained:

$$\text{dom}(\exp_a) = \mathbb{R}, \quad \text{ran}(\exp_a) = (0, \infty) = \mathbb{R}^{>0}.$$

For $a > 1$ the function a^x is strictly increasing; for $0 < a < 1$ it is strictly decreasing; for $a = 1$ it is the constant 1. In all cases $a^0 = 1$ and $a^{x+y} = a^x a^y$.

5.5 Logarithms

Definition 5.5.

Fix a base $a > 0$ with $a \neq 1$. Then $\exp_a$ is injective from $\mathbb{R}$ onto $(0, \infty)$, and its inverse function (in the sense of Definition 4.8) is the **logarithm to base a** ,

$$\log_a : (0, \infty) \rightarrow \mathbb{R}, \quad \log_a(a^x) = x, \quad a^{\log_a y} = y.$$

Thus

$$\text{dom}(\log_a) = (0, \infty), \quad \text{ran}(\log_a) = \mathbb{R}.$$

Because it is the inverse of a^x , the graph of $\log_a$ is the reflection of the graph of a^x across the line $y = x$ (Remark 4.3). It converts products to sums: $\log_a(xy) = \log_a x + \log_a y$.

5.6 Trigonometric Functions

Let x be a real number, thought of as an angle measured in **radians**. Rotating the point $(1, 0)$ counter-clockwise through the angle x about the origin lands on a point of the unit circle whose coordinates are, by definition, $(\cos x, \sin x)$.

Both $\sin$ and $\cos$ are defined for all $x \in \mathbb{R}$, take values in $[-1, 1]$, and repeat after a full turn.

Definition 5.6.

A function f is **periodic with period** $P > 0$ if $f(x + P) = f(x)$ for every x in its domain.

Thus $\sin$ and $\cos$ are periodic with period 2π . The **tangent**

$$\tan x = \frac{\sin x}{\cos x}$$

is defined wherever $\cos x \neq 0$, i.e. for $x \neq \frac{\pi}{2} + k\pi$ with $k \in \mathbb{Z}$, and is periodic with period π .

Remark 5.3. A general **sinusoid** has the form

$$f(x) = A \sin(bx) + C,$$

with **amplitude** $|A|$ (half the vertical spread of the graph), **period** $\frac{2\pi}{b}$, and **vertical offset** C (the horizontal line the graph oscillates around).

Exercise 5.1. The function $f(x) = 3 \sin(2x) + 1$ is a sinusoid. Find its amplitude, period, and range, and the smallest $x > 0$ at which it attains its maximum.

5.7 Summary

- **Polynomials** (Definition 5.1) are finite combinations of the powers $1, x, x^2, \dots$; they are defined on all of $\mathbb{R}$.
- **Rational functions** (Definition 5.2) are quotients of polynomials, defined off the finite zero set of the denominator.
- **Power functions** x^α (Definition 5.3) have domain $\mathbb{R}$ or $\mathbb{R} \setminus \{0\}$ for integer exponents, and are typically taken on $[0, \infty)$ otherwise; together with roots they form the **algebraic** functions.
- **Exponentials** a^x (Definition 5.4) have domain $\mathbb{R}$ and range $(0, \infty)$; their inverses, the **logarithms** $\log_a$ (Definition 5.5), have domain $(0, \infty)$ and range $\mathbb{R}$.

- The **trigonometric functions** $\sin, \cos$ are 2π -periodic (Definition 5.6) with range $[-1, 1]$; $\tan = \sin / \cos$ is π -periodic and undefined where $\cos$ vanishes.

6 Continuity

Companion reading

This lecture corresponds to C&J §1.2.d.

Intuitively, a function is continuous if a small change in the independent variable produces only a small change in the dependent variable — the graph has no jumps, gaps, or blow-ups. Before making this precise, it helps to see why the intuitive idea is not yet a *definition*: it uses the word “small” twice without saying how small, or how the two smallnesses are related.

Example 6.1.

Let

$$f(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0. \end{cases}$$

At $x_0 = 0$, an arbitrarily small change in x — moving from $x = 0$ to some $x < 0$ — causes f to jump from 1 to 0, a change of a full unit. No matter how small a step we take to the left of 0, the output change stays large. We call this a **jump discontinuity** at $x_0 = 0$.

Q: How do we make “small change forces small change” precise? The idea is to formalize “small” using neighborhoods.

Definition 6.1.

Given $a \in \mathbb{R}$ and $\varepsilon > 0$, the ε -**neighborhood** of a is the interval

$$(a - \varepsilon, a + \varepsilon).$$

Think of ε as a *tolerance*: saying “ x is within tolerance of a ” means $x \in (a - \varepsilon, a + \varepsilon)$, i.e. $|x - a| < \varepsilon$. “Small change” becomes “ ε small.”

Fix a point x_0 . We want: the distance between $f(x)$ and $f(x_0)$ stays within tolerance ε , i.e.

$$|f(x) - f(x_0)| < \varepsilon,$$

whenever x is *sufficiently close* to x_0 — say, within some tolerance δ of x_0 . The key point is that this has to work for **any** $\varepsilon > 0$, no matter how small; but we are allowed to choose δ *in response to* ε — a smaller tolerance on the output may well force a smaller tolerance on the input.

6.1 The ε - δ Definition

Definition 6.2.

Let $f: D \rightarrow \mathbb{R}$ with $D \subseteq \mathbb{R}$, and let $x_0 \in D$. We say f is **continuous at** x_0 if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x \in D \text{ and } |x - x_0| < \delta \quad \Rightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

Example 6.2.

Take f as in Example 6.1 and $x_0 = 0$. Pick $\varepsilon = \frac{1}{2}$. No matter how small we choose $\delta > 0$, there is always some $x < 0$ with $|x - x_0| < \delta$ (i.e. x is δ -close to $x_0 = 0$), and for any such x ,

$$|f(x) - f(x_0)| = |0 - 1| = 1 > \frac{1}{2}.$$

So no δ works for this ε , and f is **not** continuous at $x_0 = 0$.

Remark 6.1. Continuity is a property of f **at a point**. The function f above is continuous at every $x_0 \neq 0$ (Exercise 6.1) and discontinuous only at $x_0 = 0$ — a function can be continuous at some points of its domain and discontinuous at others.

Exercise 6.1. Show that f from Example 6.1 is continuous at every $x_0 \neq 0$.

6.1.1 The ε - δ game

Remark 6.2. It helps to think of the definition as a game between an opponent and you. The opponent challenges you with some $\varepsilon > 0$; you must exhibit a matching $\delta > 0$ so that

$$|x - x_0| < \delta \quad \Rightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

f is continuous at x_0 exactly when you can win this game **no matter which** ε **the opponent picks**.

Example 6.3.

$g(x) = 3x + 2$ is continuous at every $x_0 \in \mathbb{R}$.

Proof. Let $\varepsilon > 0$ and $x_0 \in \mathbb{R}$ be arbitrary. We want $|g(x) - g(x_0)| < \varepsilon$. Compute

$$|g(x) - g(x_0)| = |(3x + 2) - (3x_0 + 2)| = |3x - 3x_0| = 3|x - x_0|.$$

So if we let $\delta = \varepsilon/3$, then

$$|x - x_0| < \delta \quad \Rightarrow \quad |g(x) - g(x_0)| = 3|x - x_0| < 3\delta = \varepsilon.$$

□

Exercise 6.2. Show, directly from Definition 6.2, that $h(x) = \frac{1}{2}x - 1$ is continuous at every $x_0 \in \mathbb{R}$. (What choice of δ in terms of ε makes the argument work?)

A Jupyter demonstration accompanying this lecture lets you play the ε - δ game interactively: pick x_0 and a δ , and watch the resulting range of f -values shrink (or fail to shrink) as $\delta \rightarrow 0$.

6.2 Types of Discontinuities

If f fails to be continuous at x_0 , the failure typically has one of three flavors.

Remark 6.3. Jump discontinuities. The values of f approach one number from the left of x_0 and a different number from the right, as in Example 6.1. We will make this precise next week with *one-sided limits*: a jump discontinuity is exactly a point where the one-sided limits exist but disagree.

Remark 6.4. Unbounded (infinite) discontinuities. Consider

$$f(x) = \begin{cases} 1/x^2 & x \neq 0 \\ 0 & x = 0. \end{cases}$$

As $x \rightarrow 0$, $f(x) \rightarrow \infty$, so $|f(x) - f(0)|$ is **unbounded** on every neighborhood of 0 — no ε -tolerance, however large, can be met near $x_0 = 0$ by any δ .

Remark 6.5. Oscillation discontinuities. Consider

$$g(x) = \begin{cases} \sin(1/x) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

Near $x_0 = 0$, g oscillates wildly: arbitrarily close to 0 we can always find points where g equals 1 and points where g equals -1 . Indeed, for any $n \in \mathbb{N}$, taking

$$x_1 = \frac{1}{2\pi n + \pi/2} \quad \text{gives} \quad \sin(1/x_1) = 1,$$

and similarly $x_{-1} = \frac{1}{2\pi n + 3\pi/2}$ gives $\sin(1/x_{-1}) = -1$; both $x_1, x_{-1} \rightarrow 0$ as $n \rightarrow \infty$. So on every neighborhood of 0, no matter how small, g takes values ε apart for ε close to 2 — g cannot be made continuous at 0 by any choice of $g(0)$.

Example 6.4.

Contrast this with

$$x \sin(1/x) \quad (x \neq 0), \quad \text{value } 0 \text{ at } x = 0,$$

which **is** continuous at $x_0 = 0$: although $\sin(1/x)$ still oscillates without settling down, it is squeezed between -1 and 1 , so $|x \sin(1/x) - 0| \leq |x| \rightarrow 0$ as $x \rightarrow 0$. Oscillation alone does not cause a discontinuity — it must fail to be *damped out* as $x \rightarrow x_0$.

6.3 Summary

- The intuitive idea “small change in, small change out” is formalized using **ε -neighborhoods** (Definition 6.1): a tolerance interval $(a - \varepsilon, a + \varepsilon)$ around a point.
- f is **continuous at** x_0 (Definition 6.2) if for every tolerance $\varepsilon > 0$ on the output, there is a matching tolerance $\delta > 0$ on the input, so that x within δ of x_0 forces $f(x)$ within ε of $f(x_0)$ — think of it as a game (Remark 6.2) you must be able to win against any ε .
- Continuity is a property of f **at a point** (Remark 6.1); a function can be continuous at some points and not others.
- Failures of continuity come in three standard flavors: **jump** discontinuities (Remark 6.3), where one-sided behavior disagrees; **unbounded** discontinuities (Remark 6.4), where f blows up; and **oscillation** discontinuities (Remark 6.5), where f never settles down near x_0 — unless the oscillation is damped, as in $x \sin(1/x)$ (Example 6.4), which remains continuous.

7 Activity: Properties of Functions

About this activity

An in-class, pencil-and-paper worksheet. It revisits the properties of functions built up in the previous chapters — *injective*, *monotone*, *even/odd*, and, above all, *continuous* — by working them out in full for two specific functions. Companion reading: C&J §1.2.d.

7.1 Background

We have met several properties a function can have: **injective**, **monotone**, **even or odd**, and — most important — **continuous**. In this activity we investigate all of them for the two functions

$$f(x) = \begin{cases} \frac{x}{|x|} & x \neq 0, \\ 0 & x = 0, \end{cases} \quad g(x) = \begin{cases} \frac{1}{x^2} & x \neq 0, \\ 0 & x = 0. \end{cases}$$

7.2 Step 1: Draw the graphs

Sketch the graph of each function on its own set of axes. Plot a few sample points first (e.g. $x = \pm\frac{1}{2}, \pm 1, \pm 2, \pm 3$), then connect them.

7.3 Step 2: Make an educated guess

From your graphs, try to answer the following.

1. Is the function injective?
2. Is it monotone? If so, is it *strictly* monotone? If not, can you find intervals — as large as possible — on which it is (strictly) monotone?
3. Is it even, odd, or neither?

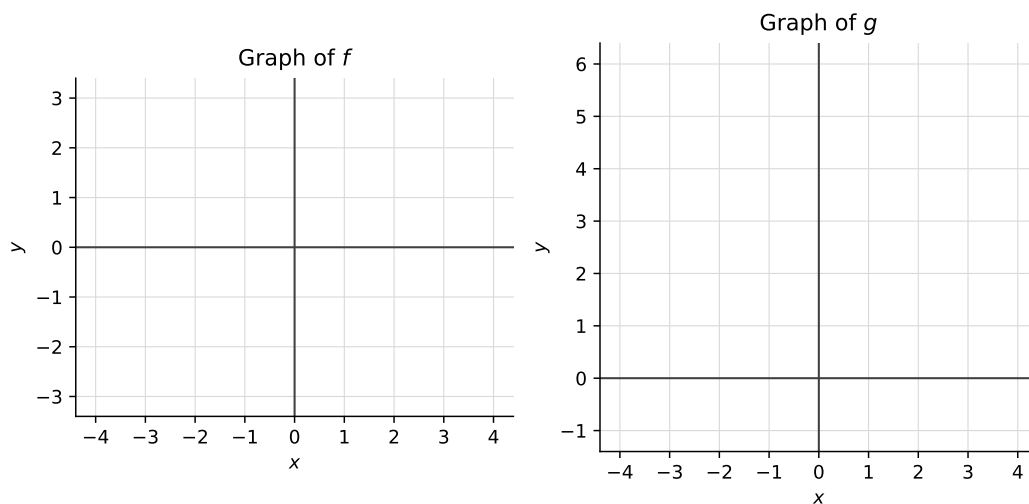


Figure 7.1: Blank axes for sketching f (left) and g (right).

4. Is it continuous everywhere? If not, at which x is it discontinuous?

Record short answers in the table; you will verify each one in the steps that follow.

Property	f	g
Injective? (yes / no)		
Monotone? Strictly? On which interval(s)?		
Even, odd, or neither?		
Continuous everywhere? If not, at which x ?		

7.4 Verification: injectivity

Neither f nor g is injective. To prove this, it is enough to exhibit two inputs $x \neq y$ with the same output.

Exercise 7.1. Find explicit values:

- for f : $x = \underline{\hspace{1cm}}$ and $y = \underline{\hspace{1cm}}$ with $f(x) = f(y)$;
- for g : $x = \underline{\hspace{1cm}}$ and $y = \underline{\hspace{1cm}}$ with $g(x) = g(y)$.

7.5 Verification: even / odd

Exercise 7.2. For whichever symmetry you guessed, verify it directly from the definition of the function: compute $f(-x)$ and check that it equals $-f(x)$ (odd) or $f(x)$ (even). Then do the same for g .

7.6 Verification: monotonicity

Exercise 7.3. Verify your monotonicity guess on a suitable interval, directly from the definition. For instance, to show a function is increasing on an interval, assume $x \leq y$ in that interval and deduce $f(x) \leq f(y)$ (use strict inequalities for *strictly* increasing); argue similarly for decreasing.

7.7 Continuity: the ε - δ game

Both f and g are continuous at $x_0 = 1$. Play the ε - δ game from Definition 6.2.

Exercise 7.4.

1. Take $\varepsilon = 1$. Find a number $\delta > 0$ such that

$$|x - 1| < \delta \quad \Rightarrow \quad |f(x) - f(1)| < 1$$

(recall $f(1) = 1$). Then find such a δ for g .

2. Repeat with a smaller tolerance, $\varepsilon = \frac{1}{10}$. Does the argument really change?
3. Can you produce a δ that works for an **arbitrary** $\varepsilon > 0$?

7.8 Discontinuities

Both functions are discontinuous at $x_0 = 0$. Here it is *you* who loses the game.

Exercise 7.5. Find an $\varepsilon > 0$ for which **no** δ works — that is, however close x is taken to 0, there is always some such x with

$$|f(x) - f(0)| \geq \varepsilon.$$

Give such an ε for f , and one for g .

Part III

Limits

8 Sequences and Convergence

8.1 Sequences

A **sequence** is an unending list of real numbers

$$a_1, a_2, a_3, \dots$$

We write $(a_n)_{n \in \mathbb{N}}$, or just (a_n) , and call a_n the **n -th term**. Formally, a sequence is nothing but a function

$$a: \mathbb{N} \rightarrow \mathbb{R},$$

where we write a_n instead of $a(n)$.

Example 8.1.

$$\begin{aligned} a_n &= \frac{1}{n} : & 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \\ b_n &= \frac{(-1)^{n+1}}{n} : & 1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots \\ c_n &= \frac{n}{n+1} : & \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots \\ d_n &= \sqrt[n]{p}, \quad p > 0 \text{ fixed} \\ e_n &= \alpha^n, \quad \alpha > 1. \end{aligned}$$

8.2 The Limit of a Sequence

Look at $a_n = \frac{1}{n}$. What happens to the terms as n gets larger and larger? The values get smaller and smaller, coming **arbitrarily close to 0** — though no term is ever equal to 0. We want to make “arbitrarily close” precise.

The idea is to use ε -neighborhoods (Definition 6.1): given a tolerance $\varepsilon > 0$, we ask that *from some point on*, every term of the sequence lies within ε of 0.

Definition 8.1.

We say (a_n) **converges to 0**, written

$$\lim_{n \rightarrow \infty} a_n = 0 \quad \text{or} \quad a_n \rightarrow 0,$$

if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_n| = |a_n - 0| < \varepsilon \quad \text{for all } n \geq N.$$

The number N is allowed to depend on ε — a smaller tolerance will in general require going further out in the sequence. To emphasize this dependence one sometimes writes $N(\varepsilon)$.

Example 8.2.

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Proof. Let $\varepsilon > 0$. We must find N so that $\frac{1}{n} < \varepsilon$ for all $n \geq N$. Now

$$\frac{1}{n} < \varepsilon \quad \Leftrightarrow \quad \frac{1}{\varepsilon} < n,$$

so any $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$ works: if $n \geq N > \frac{1}{\varepsilon}$, then $\frac{1}{n} < \varepsilon$.

Concretely, for $\varepsilon = 0.05 = \frac{5}{100}$ we need $n > \frac{100}{5} = 20$, so $N = 21$ will do. $\square$

Exercise 8.1. Show that $\lim_{n \rightarrow \infty} \frac{(-1)^{n+1}}{n} = 0$. (Note $|b_n| = \frac{1}{n}$.)

8.3 Convergence to a General Limit

Now consider $c_n = \frac{n}{n+1}$. The terms get closer and closer to 1, which suggests $\lim_{n \rightarrow \infty} c_n = 1$. The definition is the same as before, with 0 replaced by the candidate limit.

Definition 8.2.

Let $a \in \mathbb{R}$. We say (c_n) **converges to** a , written $\lim_{n \rightarrow \infty} c_n = a$ or $c_n \rightarrow a$, if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|c_n - a| < \varepsilon \quad \text{for all } n \geq N.$$

Equivalently: from the N -th term on, all terms lie in the ε -neighborhood $(a - \varepsilon, a + \varepsilon)$. A sequence is **convergent** if it converges to some a , and **divergent** otherwise.

Example 8.3.

$\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$. Here

$$|c_n - 1| = \left| \frac{n}{n+1} - 1 \right| = 1 - \frac{n}{n+1} = \frac{1}{n+1}.$$

Let us check two tolerances. For $\varepsilon = 1$ we need $\frac{1}{n+1} < 1$, which holds for every $n \geq 1$; so $N = 1$ works. For $\varepsilon = \frac{1}{4}$ we need

$$\frac{1}{n+1} < \frac{1}{4} \quad \Leftrightarrow \quad 3 < n,$$

so $N = 4$ works.

In general, $|c_n - 1| = \frac{1}{n+1} < \varepsilon \quad \Leftrightarrow \quad n > \frac{1}{\varepsilon} - 1$, so any $N > \frac{1}{\varepsilon} - 1$ works, and $c_n \rightarrow 1$.

Remark 8.1. When a limit exists it is **unique**: if $c_n \rightarrow a$ and $c_n \rightarrow a'$, then for every $\varepsilon > 0$ we have $|a - a'| \leq |a - c_n| + |c_n - a'| < 2\varepsilon$ for large n , so $|a - a'| = 0$. This is what justifies speaking of *the* limit and the notation $\lim_{n \rightarrow \infty} c_n$.

8.3.1 Divergence

To show a sequence **diverges** we must show it converges to no limit at all. Negating Definition 8.2: (c_n) does not converge to a if there is some $\varepsilon > 0$ such that, no matter how far out we go, some later term is at least ε away from a .

Example 8.4.

The sequence $((-1)^n) = -1, 1, -1, 1, \dots$ diverges.

Proof. Suppose $(-1)^n \rightarrow a$ for some a . Take $\varepsilon = 1$. Then $|(-1)^n - a| < 1$ for all large n ; applying this to an even and an odd n gives $|1 - a| < 1$ and $|-1 - a| < 1$. But then

$$2 = |1 - (-1)| \leq |1 - a| + |a - (-1)| < 1 + 1 = 2,$$

a contradiction. □

Exercise 8.2. Investigate the remaining sequences from Example 8.1: $d_n = \sqrt[n]{p}$ (with $p > 0$ fixed) and $e_n = \alpha^n$ (with $\alpha > 1$). One converges; the other diverges. Which is which, and what is the limit?

Remark 8.2. For reference: $\sqrt[n]{p} \rightarrow 1$ for every $p > 0$, while α^n diverges when $\alpha > 1$ (its terms grow without bound). Both facts are proved in C&J §1.6.

8.4 Limit Laws

Computing $N(\varepsilon)$ by hand for every sequence quickly becomes tedious. The following rules let us build limits of complicated sequences out of simpler ones.

Proposition 8.1 (Arithmetic of limits). *Suppose $a_n \rightarrow a$ and $b_n \rightarrow b$. Then*

$$\begin{aligned} a_n + b_n &\rightarrow a + b, & a_n - b_n &\rightarrow a - b, \\ a_n \cdot b_n &\rightarrow a \cdot b, & \frac{a_n}{b_n} &\rightarrow \frac{a}{b} \quad (\text{provided } b \neq 0). \end{aligned}$$

Example 8.5.

Find $\lim_{n \rightarrow \infty} \frac{2n^2 - 1}{n^2 + n + 1}$.

Divide numerator and denominator by n^2 :

$$\frac{2n^2 - 1}{n^2 + n + 1} = \frac{2 - \frac{1}{n^2}}{1 + \frac{1}{n} + \frac{1}{n^2}}.$$

Now $\frac{1}{n} \rightarrow 0$ and $\frac{1}{n^2} \rightarrow 0$, so by the limit laws the numerator tends to $2 - 0 = 2$ and the denominator to $1 + 0 + 0 = 1$. Since the denominator limit is nonzero,

$$\lim_{n \rightarrow \infty} \frac{2n^2 - 1}{n^2 + n + 1} = \frac{2}{1} = 2.$$

8.5 Comparison and the Squeeze Theorem

Exercise 8.3. Warm-up. Show that $\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$.

Rather than hunting for $N(\varepsilon)$ from scratch every time, it is often easier to compare a sequence to one whose limit we already know.

Proposition 8.2 (Comparison test). *Suppose $0 \leq b_n \leq a_n$ for all n , and $a_n \rightarrow 0$. Then $b_n \rightarrow 0$.*

Proof. Let $\varepsilon > 0$. Since $a_n \rightarrow 0$ there is $N \in \mathbb{N}$ with $a_n < \varepsilon$ for all $n \geq N$. Then for $n \geq N$,

$$0 \leq b_n \leq a_n < \varepsilon,$$

so $|b_n - 0| = b_n < \varepsilon$. Hence $b_n \rightarrow 0$. □

We can generalize this by sandwiching a sequence between two others that converge to the *same* limit works for any target value.

Proposition 8.3 (Squeeze theorem). *Suppose $a_n \leq b_n \leq c_n$ for all n , and $a_n \rightarrow L$, $c_n \rightarrow L$ for the same $L \in \mathbb{R}$. Then $b_n \rightarrow L$.*

Proof. Let $\varepsilon > 0$. Choose N_1 with $|a_n - L| < \varepsilon$ for $n \geq N_1$, and N_2 with $|c_n - L| < \varepsilon$ for $n \geq N_2$. Let $N = \max(N_1, N_2)$. For $n \geq N$,

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon,$$

so $|b_n - L| < \varepsilon$. Hence $b_n \rightarrow L$. $\square$

Remark 8.3. Proposition 8.2 is the special case of the squeeze theorem where the lower bounding sequence is constantly 0 and $L = 0$.

8.6 Monotone and Bounded Sequences

Definition 8.3.

A sequence (a_n) is **increasing** if $a_n \leq a_{n+1}$ for all n , and **decreasing** if $a_n \geq a_{n+1}$ for all n . It is **monotone** if it is increasing or decreasing.

Definition 8.4.

A sequence (a_n) is **bounded** if there is a constant C such that $|a_n| \leq C$ for all n .

Theorem 8.1 (Monotone Convergence Theorem). *Every bounded monotone sequence converges.*

Remark 8.4. This theorem expresses the same completeness of $\mathbb{R}$ as the Nested Interval Axiom from [The Nested Interval Axiom](#) — each can be derived from the other. We will not reprove the general statement here; instead we isolate the paired form that we actually use below, which follows directly from the axiom together with the comparison test.

Corollary 8.1. *Suppose (a_n) is increasing, (b_n) is decreasing, $a_n \leq b_n$ for all n , and $b_n - a_n \rightarrow 0$. Then (a_n) and (b_n) converge to a common limit c , the unique point in $\bigcap_n [a_n, b_n]$.*

Proof. Since (a_n) is increasing and (b_n) is decreasing with $a_n \leq b_n$, the intervals $[a_n, b_n]$ are nested: $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ for all n . By the Nested Interval Axiom there is a point c with $a_n \leq c \leq b_n$ for every n . Then

$$0 \leq c - a_n \leq b_n - a_n \rightarrow 0 \quad \text{and} \quad 0 \leq b_n - c \leq b_n - a_n \rightarrow 0,$$

so the comparison test (Proposition 8.2) gives $a_n \rightarrow c$ and $b_n \rightarrow c$. $\square$

8.7 Applications

Lemma 8.1 (Bernoulli's inequality). *For $c \geq -1$ and $n \in \mathbb{N}$,*

$$(1 + c)^n \geq 1 + nc.$$

(For $c \geq 0$ one can prove this by considering just the first two terms of the [binomial expansion](#) of $(1 + c)^n$. The remaining terms are all non-negative, so by leaving them out, we obtain something smaller.)

8.7.1 Roots: $\sqrt[n]{p} \rightarrow 1$

Theorem 8.2.

For every $p > 0$, $\lim_{n \rightarrow \infty} \sqrt[n]{p} = 1$.

Proof. For $p = 1$ the sequence is constant, so there is nothing to show.

Suppose $p > 1$. Write $a_n := \sqrt[n]{p} = 1 + \delta_n$; since $p > 1$ we have $a_n > 1$, so $\delta_n > 0$. Raising to the n -th power and applying Bernoulli's inequality (with $c = \delta_n \geq 0$),

$$p = (1 + \delta_n)^n \geq 1 + n\delta_n,$$

so $0 < \delta_n \leq \frac{p-1}{n}$. Since $\frac{p-1}{n} \rightarrow 0$, the comparison test gives $\delta_n \rightarrow 0$, and hence $a_n = 1 + \delta_n \rightarrow 1$.

Suppose $0 < p < 1$. Then $q := 1/p > 1$, and by the case just proved $\sqrt[n]{q} \rightarrow 1$. Since $\sqrt[n]{q} \neq 0$ for all n and its limit is nonzero, the quotient rule in [Proposition 8.1](#) gives

$$\sqrt[n]{p} = \frac{1}{\sqrt[n]{q}} \rightarrow \frac{1}{1} = 1.$$

□

This proves the claim recorded in [Remark 8.2](#) for $d_n = \sqrt[n]{p}$.

8.7.2 The Number e

Proposition 8.4.

The sequence $e_n := \left(1 + \frac{1}{n}\right)^n$ is increasing.

Proof. We must show

$$\left(1 + \frac{1}{n}\right)^n \leq \left(1 + \frac{1}{n+1}\right)^{n+1}. \quad (*)$$

Writing both sides with a common fractional base and splitting off one factor of $\frac{n+2}{n+1}$ on the right,

$$(*) \iff \left(\frac{n+1}{n}\right)^n \leq \left(\frac{n+2}{n+1}\right)^n \cdot \frac{n+2}{n+1} \iff \frac{n+1}{n+2} \leq \left(\frac{n(n+2)}{(n+1)^2}\right)^n.$$

Since $(n+1)^2 = n^2 + 2n + 1$, we have $n(n+2) = n^2 + 2n = (n+1)^2 - 1$, so

$$\left(\frac{n(n+2)}{(n+1)^2}\right)^n = \left(1 - \frac{1}{(n+1)^2}\right)^n.$$

By Bernoulli's inequality (with $c = -\frac{1}{(n+1)^2} \geq -1$),

$$\left(1 - \frac{1}{(n+1)^2}\right)^n \geq 1 - \frac{n}{(n+1)^2}.$$

So it suffices to show $\frac{n+1}{n+2} \leq 1 - \frac{n}{(n+1)^2}$, i.e.

$$\frac{1}{n+2} \geq \frac{n}{(n+1)^2} \iff (n+1)^2 \geq n(n+2) \iff n^2 + 2n + 1 \geq n^2 + 2n,$$

which is always true. Chaining the inequalities back together proves (*). $\square$

Exercise 8.4. Show that $f_n := \left(1 + \frac{1}{n}\right)^{n+1}$ is decreasing. (Adapt the computation in the proof of Proposition 8.4.)

Theorem 8.3.

The limit

$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

exists; numerically $e \approx 2.71828 \dots$

Proof. Let $e_n = (1 + \frac{1}{n})^n$ and $f_n = (1 + \frac{1}{n})^{n+1} = e_n (1 + \frac{1}{n})$, so $e_n \leq f_n$ for all n . By Proposition 8.4, (e_n) is increasing; by Exercise 8.4, (f_n) is decreasing. In particular $e_n \leq f_n \leq f_1 = 4$ for all n , so (e_n) is bounded.

Moreover

$$0 \leq f_n - e_n = e_n \cdot \frac{1}{n} \leq \frac{f_1}{n} = \frac{4}{n} \rightarrow 0,$$

so $f_n - e_n \rightarrow 0$ by the comparison test. By Corollary 8.1, (e_n) and (f_n) converge to a common limit — which we call e . Numerically, $e_{10} \approx 2.594$, $e_{100} \approx 2.705$, $e_{1000} \approx 2.7169$, consistent with $e \approx 2.71828 \dots$ $\square$

8.8 Summary

- A **sequence** is a function $a: \mathbb{N} \rightarrow \mathbb{R}$, written (a_n) .
- (a_n) **converges to** a (Definition 8.2) if for every tolerance $\varepsilon > 0$ there is an index N such that $|a_n - a| < \varepsilon$ for all $n \geq N$ — from some point on, all terms lie in the ε -neighborhood of a . The special case $a = 0$ is Definition 8.1.
- The limit, when it exists, is **unique** (Remark 8.1). A sequence with no limit, such as $((-1)^n)$ (Example 8.4), is **divergent**.
- The **limit laws** (Proposition 8.1) compute limits of sums, differences, products, and quotients termwise; dividing through by the highest power of n turns a ratio of polynomials into a form where they apply (Example 8.5).
- The **comparison test** and **squeeze theorem** (Proposition 8.2, Proposition 8.3) let us find a limit by trapping a sequence between two others whose limits we already know.
- A **bounded monotone sequence converges** (2), a fact equivalent to the Nested Interval Axiom; the paired form (Corollary 8.1) — an increasing and a decreasing sequence squeezing together — is the version we use in practice.
- **Bernoulli's inequality** $(1 + c)^n \geq 1 + nc$ (Lemma 8.1) drives two applications: every $\sqrt[n]{p} \rightarrow 1$ (Theorem 8.2), and $(1 + \frac{1}{n})^n$ increases while $(1 + \frac{1}{n})^{n+1}$ decreases to the same limit, defining the number $e \approx 2.71828 \dots$ (Theorem 8.3).

9 Convergence of Sequences: Reading Data

```
#
# !!! RUN THIS FIRST !!!
#
# (loads the data and defines all necessary functions)
#

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

pd.set_option('display.max_rows', 20) # long tables show head/tail with '...' automatic

df = pd.read_csv('data/rc_circuit_data.csv')

def a(n):
    """Return the n-th term of the sequence, a_n, as a plain float."""
    row = df.loc[df['n'] == n]
    if row.empty:
        raise ValueError(f"n = {n} is out of range (data covers n = 0 .. {df['n'].max()})")
    return float(row['measured_value'].values[0])

def find_smallest_N(data, L_guess, epsilon, value_col='measured_value'):
    """Smallest N (present in the data) such that all later points are within epsilon of
    vals = data[value_col].values
    n_vals = data['n'].values
    for i in range(len(vals)):
        if np.all(np.abs(vals[i:] - L_guess) < epsilon):
            return int(n_vals[i])
    return None
```

```

def check_convergence(data, L_guess, N_guess, epsilon, value_col='measured_value', reveal=True):
    vals = data[value_col].values
    n_vals = data['n'].values
    mask = n_vals >= N_guess

    if not mask.any():
        print(f"N_guess = {N_guess} is beyond the range of available data (max n = {n_vals[-1]})")
        return False

    tail_n = n_vals[mask]
    tail_vals = vals[mask]
    bad = np.abs(tail_vals - L_guess) >= epsilon

    if bad.any():
        first_bad_n = tail_n[bad][0]
        print(f" Not valid: at n = {first_bad_n}, |a_n - {L_guess}| >= {epsilon}.")
        print(" The sequence leaves the epsilon-band after your N_guess. Try a larger N_guess "
              "or reconsider your L_guess.")
        return False

    print(f" Valid: for all n >= {N_guess}, |a_n - {L_guess}| < {epsilon}.")

    # Don't just hand over the smallest N -- only reveal it once the guess is close.
    best_N = find_smallest_N(data, L_guess, epsilon, value_col)
    if best_N is not None:
        gap = N_guess - best_N
        if gap == 0:
            print(" This is also the smallest N that works for this L_guess.")
        elif gap <= reveal_threshold:
            print(f" Close! The smallest N that works is N = {best_N} (you were only {gap} away)")
        else:
            print(" This N works, but it's probably not the smallest one -- "
                  "try a smaller N and see if it still holds.")
    return True

def plot_challenge(data, L_guess, N_guess=None, epsilon=None, value_col='measured_value',
                  zoom=True, buffer=50):
    """
    Plot the sequence with the guessed L (and epsilon band, and N_guess) marked.

    By default (zoom=True), the plot starts `buffer` samples before N_guess and

```

```

runs to the end of the data -- enough to see whether the sequence was
outside the band right before N_guess, and to see it stay inside the band
all the way to the end. Set zoom=False to see the whole sequence instead,
or pass a bigger `buffer` to look further back.
"""
plot_data = data
if zoom and N_guess is not None:
    start = max(int(data['n'].min()), N_guess - buffer)
    mask = data['n'] >= start
    plot_data = data[mask]

plt.figure(figsize=(8, 4))
plt.plot(plot_data['n'], plot_data[value_col], '.', alpha=0.6)
plt.axhline(L_guess, color='green', linestyle='--', label=f'L_guess = {L_guess}')
if epsilon is not None:
    plt.fill_between(plot_data['n'], L_guess - epsilon, L_guess + epsilon,
                     color='green', alpha=0.15, label=f'epsilon band (+/- {epsilon})')
if N_guess is not None:
    plt.axvline(N_guess, color='red', linestyle=':', label=f'N_guess = {N_guess}')
plt.xlabel('n')
plt.ylabel(value_col)
plt.legend()
plt.show()

```

You've been handed a log file, `data/rc_circuit_data.csv`, from a lab sensor. It records a voltage measurement `measured_value` at each sample index `n` (and the corresponding time `t_seconds`).

You do not know what circuit or process produced this data. Your job is to:

1. Explore the data and form a hypothesis about its limiting behavior.
2. Given a challenge value ε , **guess** the limit L and the index N after which the data stays within ε of L — then verify your guess rigorously, not just by eye.

Recall the definition:

$$\lim_{n \rightarrow \infty} a_n = L \iff \text{for all } \varepsilon > 0, \text{ there is } N \text{ s.t. } n \geq N \implies |a_n - L| < \varepsilon.$$

9.1 A Simple Query Function

Sometimes you just want a single term of the sequence, a_n , without pulling up the whole table. `a(n)` does that.

```
# try it:  
  
a(0), a(5000), a(9999)
```

```
(0.0, 5.0165, 4.9733)
```

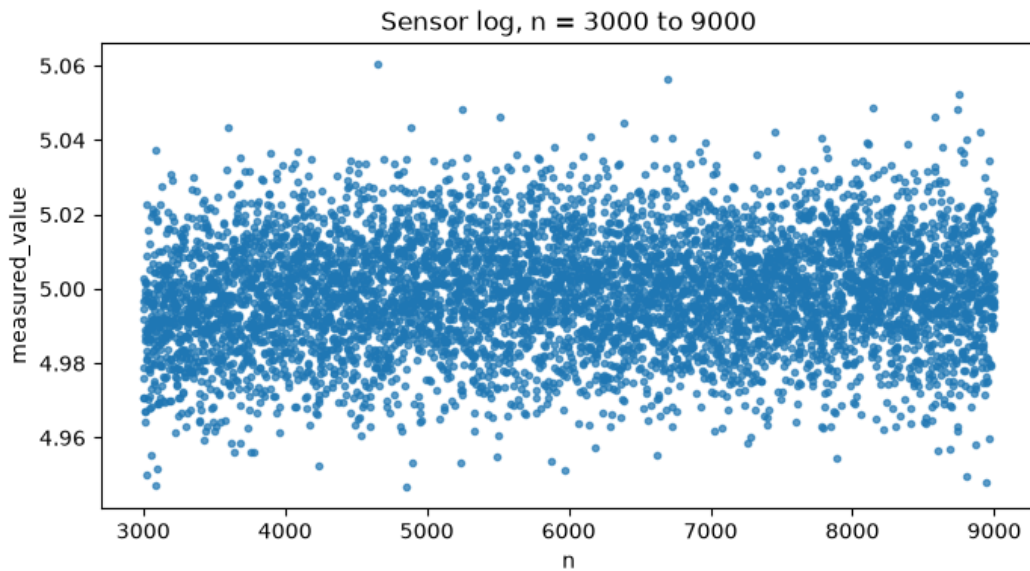
9.2 Task 1: Explore the Data

The full log has 10,000 samples. Set `n_start` and `n_end` below to pick a window and re-run the cell to see the table and plot over just that range — e.g. zoom into the early samples to see the initial behavior, or into the tail to see whether it's still moving. Change the numbers and re-run as many times as you like.

```
# EDIT HERE  
  
n_start = 3000  
n_end = 9000  
  
#  
# No need to change anything down here  
#  
view = df[(df['n'] >= n_start) & (df['n'] <= n_end)]  
display(view)  
  
plt.figure(figsize=(8, 4))  
plt.plot(view['n'], view['measured_value'], '.', alpha=0.7)  
plt.xlabel('n')  
plt.ylabel('measured_value')  
plt.title(f'Sensor log, n = {n_start} to {n_end}')  
plt.show()
```

	n	t_seconds	measured_value
3000	3000	0.3000	4.9917

	n	t_seconds	measured_value
	3001	0.3001	4.9966
	3002	0.3002	4.9946
	3003	0.3003	4.9709
	3004	0.3004	5.0027
...	...	...	...
	8996	0.8996	5.0257
	8997	0.8997	4.9995
	8998	0.8998	4.9907
	8999	0.8999	4.9952
	9000	0.9000	5.0096



9.2.1 Questions to answer before moving on

- What value does the sequence appear to be approaching? Write down your best guess for L .
- Roughly how many samples does it take before the data “settles down” near that value?
- Does the data ever exceed your guessed L , or only approach it from below?
- Try $a(n)$ for a few large n (e.g. 7000, 8500, 9999) — does it match what the plot suggested?

9.3 Task 2: The ε Challenge

Your instructor assigns a value of ε . Using **only the data** (no formulas, no ground truth), you must:

1. State your guessed limit `L_guess`.
2. State the smallest index `N_guess` you believe works.
3. Check your answer with `check_convergence` below, which verifies directly against the definition: it checks whether **every** remaining data point from `N_guess` onward is within `epsilon` of your `L_guess`.

Note this checks self-consistency with your own guessed L — it does not tell you the “true” limit, because the whole point is that you don’t get to see that. Two different reasonable guesses for L can each have a valid N .

The checker also won’t just hand you the smallest possible N : if your guess is close (within 50), it’ll tell you the exact smallest N ; if you’re further off, it’ll just tell you to try smaller.

`plot_challenge` zooms in by default: it starts a bit before your `N_guess` and runs to the end of the data, so you can see both whether the sequence was outside the band just before `N_guess` and that it stays inside all the way to the end. Pass `zoom=False` if you want to see the whole sequence instead.

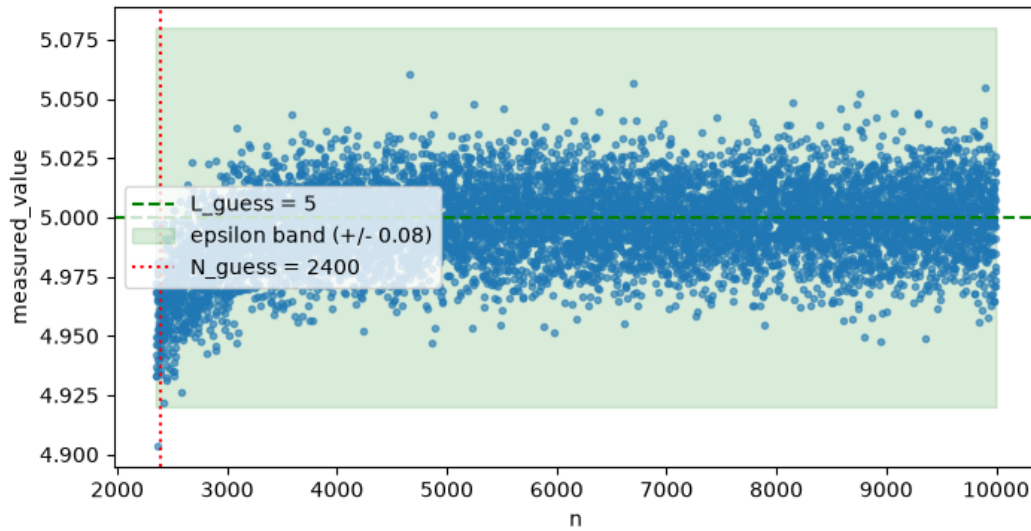
```
# --- Your answer ---
EPSILON = 0.08          # <- instructor-assigned challenge value

L_guess = 5             # TODO: your guessed limit
N_guess = 2400          # TODO: your guessed N

check_convergence(df, L_guess, N_guess, EPSILON)
plot_challenge(df, L_guess, N_guess, EPSILON)
```

Valid: for all $n \geq 2400$, $|a_n - 5| < 0.08$.

Close! The smallest N that works is $N = 2361$ (you were only 39 away).



9.4 Dataset 2: Damped Oscillator

This time there’s no CSV and no noise. Instead, you’re given the sequence directly as a **function** you can call for any index **n** — exactly the way a sequence (b_n) is usually presented in a calculus course. It models the position of an underdamped mechanical system (e.g. a suspension, a sensor needle) responding to a step input:

$$b_n = x_{ss} \left(1 - e^{-\zeta \omega_n t_n} \left(\cos(\omega_d t_n) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(\omega_d t_n) \right) \right), \quad t_n = n \cdot dt$$

with $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. Unlike Dataset 1, this sequence **oscillates** around its limit before settling — so “getting close once” is not enough; you have to check it *stays* close.

Since it’s an exact formula (not a fixed table), you can evaluate it at *any* **n** you like, no matter how large — there’s no upper limit like there was with **a(n)**.

```
x_ss = 2.0      # steady-state value
zeta = 0.1      # damping ratio
wn = 10.0      # natural frequency (rad/s)
wd = wn * np.sqrt(1 - zeta**2)
dt2 = 0.001    # seconds per sample
```

```
def b(n):
    """Return the n-th term of the damped step-response sequence, b_n, as a
    plain float (or array of floats if n is array-like). No noise: this is
    the exact closed-form value."""
    scalar_input = np.ndim(n) == 0
    t = np.atleast_1d(np.asarray(n, dtype=float)) * dt2
    vals = x_ss * (1 - np.exp(-zeta * wn * t) *
                  (np.cos(wd * t) + (zeta / np.sqrt(1 - zeta**2)) * np.sin(wd * t)))
    return float(vals[0]) if scalar_input else vals

# try it -- note n can be far larger than anything in Dataset 1's table:
b(0), b(3000), b(1_000_000)
```

(0.0, 2.0095601007352615, 2.0)

9.5 Task 3: Explore the Sequence

Same idea as Task 1: set `n_start` and `n_end` and re-run to see a table and plot of b_n over that window. Try zooming into the first couple thousand terms to see the oscillation, then look further out to see it settle.

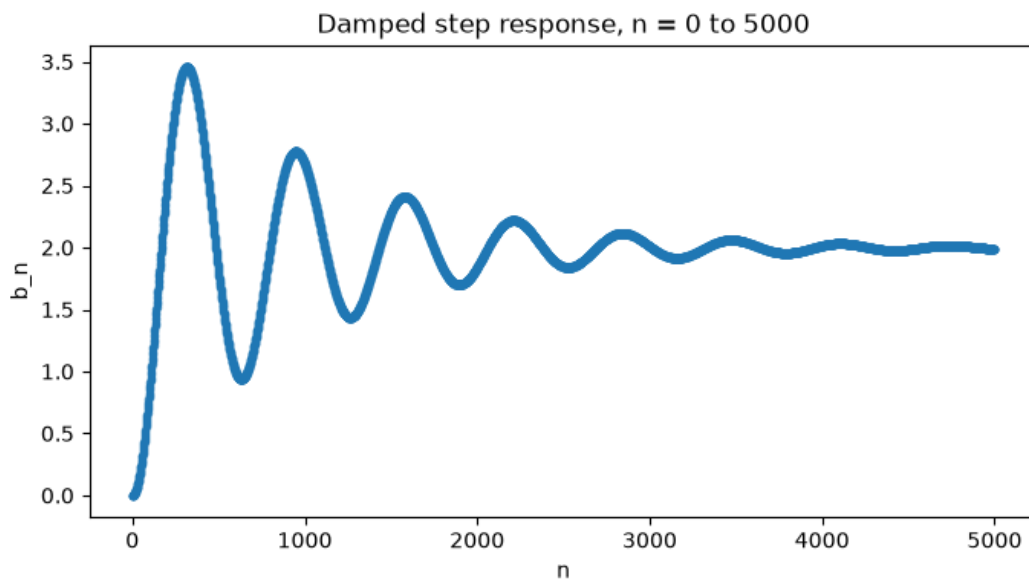
```
n_start = 0
n_end = 5000

n_vals = np.arange(n_start, n_end + 1)
view2 = pd.DataFrame({'n': n_vals, 'b_n': b(n_vals)})
display(view2)

plt.figure(figsize=(8, 4))
plt.plot(view2['n'], view2['b_n'], '.', alpha=0.7)
plt.xlabel('n')
plt.ylabel('b_n')
plt.title(f'Damped step response, n = {n_start} to {n_end}')
plt.show()
```

	n	b_n
0	0	0.000000
1	1	0.000100
2	2	0.000399

	n	b_n
3	3	0.000898
4	4	0.001596
...	...	...
4996	4996	1.989225
4997	4997	1.989154
4998	4998	1.989084
4999	4999	1.989015
5000	5000	1.988948



9.5.1 Questions to answer before moving on

- What value does b_n appear to be converging to? Compare to your guess for Dataset 1 — how did you decide this time, given there's no noise to average out?
- Around how many terms does it take before the oscillation dies down to something you'd call “close” to the limit?
- Pick an n where the plot looks close to the limit, then check a later n — does it ever swing back out? Why might that matter for choosing N ?

9.6 Task 4: The ε Challenge, Again

Same process as Task 2 — guess L_{guess} and N_{guess} for a given ε , and check them. The `check_convergence`, `find_smallest_N`, and `plot_challenge` functions from Task 2 work unchanged here; we just need a table of b_n values to check against. Since b_n is exact (no noise), we can build that table out to a very large n — far past where anything interesting happens — so the check is effectively verifying “forever.”

```
n_check = np.arange(0, 20000)      # generous horizon -- well past where b_n settles
osc_df = pd.DataFrame({'n': n_check, 'b_n': b(n_check)})

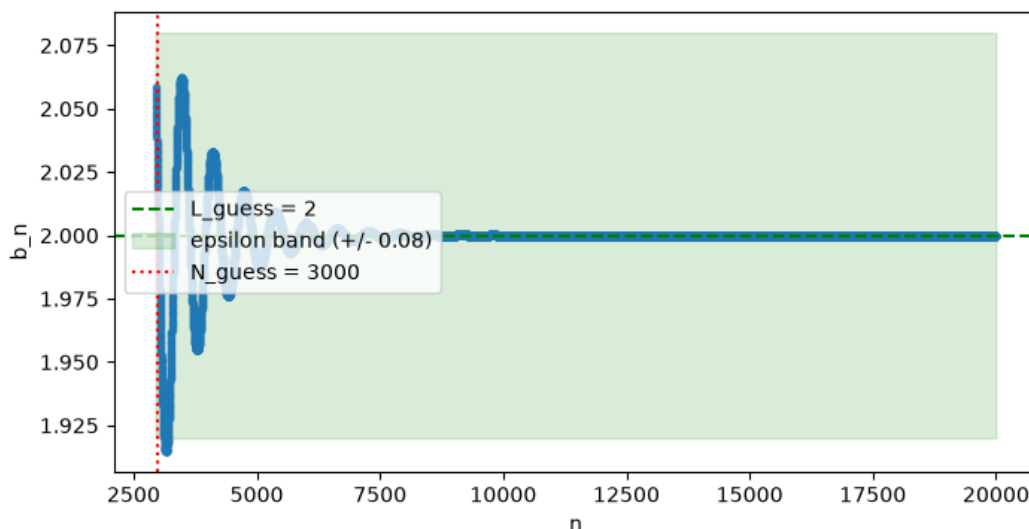
# --- Your answer ---
EPSILON2 = 0.08                    # <- instructor-assigned challenge value

L_guess2 = 2                       # TODO: your guessed limit
N_guess2 = 3000                    # TODO: your guessed N

check_convergence(osc_df, L_guess2, N_guess2, EPSILON2, value_col='b_n')
plot_challenge(osc_df, L_guess2, N_guess2, EPSILON2, value_col='b_n')
```

Not valid: at $n = 3124$, $|a_n - 2| \geq 0.08$.

The sequence leaves the epsilon-band after your N_{guess} . Try a larger N , or reconsider



9.7 Task 5: Convergence of $\sqrt[n]{p}$

Finally, we want to study the convergence behavior of $d_n = \sqrt[n]{p}$ for various p . Set the parameters in the cell below.

```
# Set p
p = 2      # Experiment with different p

# Set the range of n you want to consider
d_start = 1
d_end = 100

def d(n):
    return float(p**(1/n))
```

As above, to return just a single term of the sequence, use `d(n)`:

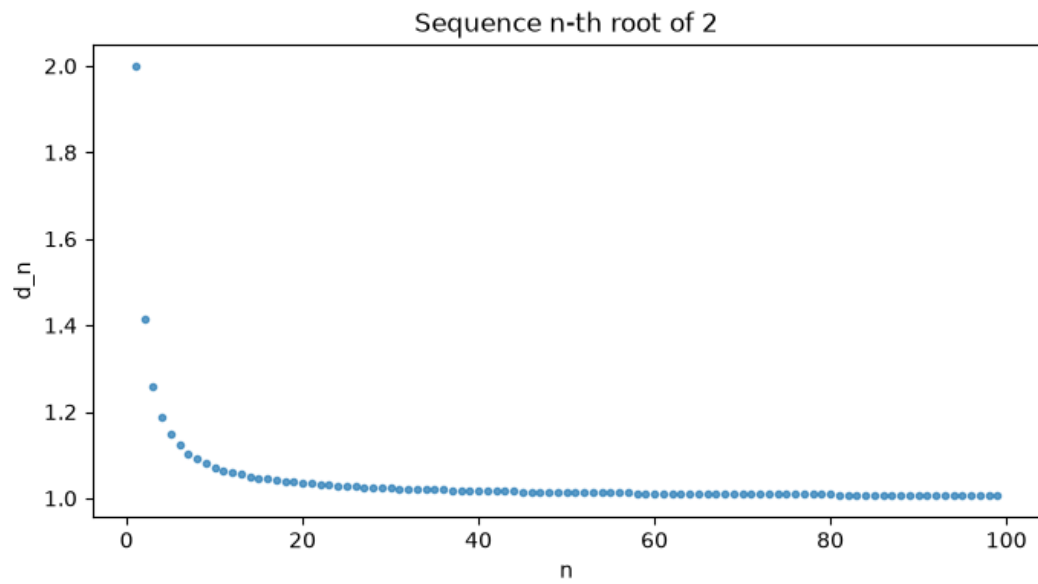
```
d(2)
```

```
1.4142135623730951
```

Now let's plot the sequence:

```
d_array = np.arange(d_start, d_end)
d_vals = p**(1/d_array)
view3 = pd.DataFrame({'n': d_array, 'd_n': d_vals})
# display(view3)

plt.figure(figsize=(8, 4))
plt.plot(view3['n'], view3['d_n'], '.', alpha=0.7)
plt.xlabel('n')
plt.ylabel('d_n')
plt.title(f'Sequence n-th root of {p}')
plt.show()
```



Questions: - Does this sequence converge for all $p > 0$? - If so, does the limit depend on p ?

10 Limits of Functions

Companion reading

This lecture corresponds to C&J §1.8 (the concept of limit for a function of a continuous variable), tying together §1.2.d (continuity) with the Intermediate Value Theorem of §1.2.e.

10.1 From Sequences to Functions

We already know what it means for a *sequence* to converge (Definition 8.2). We now transfer this idea to functions: instead of asking what happens to a_n as $n \rightarrow \infty$, we ask what happens to $f(x)$ as x approaches some fixed point a . The cheapest way to make this precise is to reduce it to something we already understand — sequences — by feeding *every* sequence approaching a into f and demanding that the outputs all converge to the same value.

Definition 10.1.

Let $f: D \rightarrow \mathbb{R}$ with $D \subseteq \mathbb{R}$, and let $a \in \mathbb{R}$. We say

$$\lim_{x \rightarrow a} f(x) = L$$

if for **every** sequence (x_n) with $x_n \in D$, $x_n \neq a$, and $x_n \rightarrow a$, we have $f(x_n) \rightarrow L$.

Remark 10.1. The word “every” is doing all the work: it is not enough to find *some* sequence $x_n \rightarrow a$ with $f(x_n) \rightarrow L$ — the same L must come out no matter which approaching sequence we feed in. This mirrors uniqueness of sequential limits (Remark 8.1): if two different sequences $x_n \rightarrow a$ and $x'_n \rightarrow a$ produced different outputs $f(x_n) \rightarrow L$ and $f(x'_n) \rightarrow L' \neq L$, the limit $\lim_{x \rightarrow a} f(x)$ would simply not exist.

Example 10.1.

Let $f(x) = x^2$. Then $\lim_{x \rightarrow 0} f(x) = 0$.

Proof. Let (x_n) be an arbitrary sequence with $x_n \rightarrow 0$. We may assume $|x_n| \leq 1$ for all n (true eventually, since $x_n \rightarrow 0$). Then

$$0 \leq f(x_n) = x_n^2 \leq |x_n|,$$

and $|x_n| \rightarrow 0$, so by the comparison test (Proposition 8.2), $x_n^2 \rightarrow 0$. Since (x_n) was an arbitrary sequence converging to 0, Definition 10.1 gives $\lim_{x \rightarrow 0} f(x) = 0$. $\square$

10.2 The ε - δ Version

Just as with continuity (Definition 6.2), the sequential definition of a function limit has an equivalent ε - δ formulation.

Proposition 10.1 (ε - δ characterization of a limit). *Let $f: D \rightarrow \mathbb{R}$, $a \in \mathbb{R}$, $L \in \mathbb{R}$. Then*

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if

for every $\varepsilon > 0$ there exists $\delta > 0$ s.t. $x \in D$, $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

The proof is the same idea as the equivalence used implicitly for continuity: if the ε - δ condition holds and $x_n \rightarrow a$, choosing N so that $|x_n - a| < \delta$ for $n \geq N$ forces $|f(x_n) - L| < \varepsilon$ for $n \geq N$, i.e. $f(x_n) \rightarrow L$. Conversely, if the ε - δ condition *fails* for some $\varepsilon > 0$, then for every $\delta = \frac{1}{n}$ we can pick a point x_n with $0 < |x_n - a| < \frac{1}{n}$ but $|f(x_n) - L| \geq \varepsilon$ — producing a sequence $x_n \rightarrow a$ with $f(x_n) \not\rightarrow L$, so the sequential definition fails too.

The only difference from Definition 6.2 is the punctured condition $0 < |x - a|$: a limit does not care what happens (or fails to happen) *at* a itself, only nearby. This is what lets us connect the two notions.

Proposition 10.2 (Continuity, restated). *Let $f: D \rightarrow \mathbb{R}$ and $a \in D$. Then f is continuous at a (Definition 6.2) if and only if*

$$\lim_{x \rightarrow a} f(x) \text{ exists and equals } f(a).$$

This is exactly Definition 6.2 with the roles of x and a relabeled: continuity says $|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$, which is the ε - δ condition of Proposition 10.1 with $L = f(a)$ (the point $x = a$ itself trivially satisfies $|f(x) - f(a)| = 0 < \varepsilon$, so punctured and unpunctured versions agree once $L = f(a)$).

10.3 Examples

Example 10.2.

$f(x) = x^3$ is continuous at every $a \in \mathbb{R}$.

Proof. Let $x_n \rightarrow a$. By the limit laws for sequences (Proposition 8.1),

$$x_n^3 = x_n \cdot x_n \cdot x_n \longrightarrow a \cdot a \cdot a = a^3.$$

Hence $f(x_n) \rightarrow f(a)$ for every sequence $x_n \rightarrow a$, so $\lim_{x \rightarrow a} f(x) = f(a)$, and Proposition 10.2 gives continuity at a . $\square$

Example 10.3.

The **Dirichlet function**

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$$

is discontinuous at every $a \in \mathbb{R}$.

Take $a = 1$. Consider the two sequences

$$x_n = 1 + \frac{1}{n} \quad (\text{rational}), \quad y_n = 1 + \frac{\sqrt{2}}{n} \quad (\text{irrational}).$$

Both converge to 1, but

$$f(x_n) \equiv 1 \rightarrow 1, \quad f(y_n) \equiv 0 \rightarrow 0 \neq f(1) = 1.$$

Since two sequences converging to $a = 1$ produce different limits for f , $\lim_{x \rightarrow 1} f(x)$ does not exist (Remark 10.1), so f is not continuous at 1.

Exercise 10.1. Adapt the argument of Example 10.3 to an arbitrary $a \in \mathbb{R}$: exhibit a rational sequence and an irrational sequence, both converging to a , on which the Dirichlet function takes different limiting values. Conclude that f is discontinuous at *every* $a \in \mathbb{R}$.

Example 10.4.

Let

$$f(x) = \begin{cases} x^3 + 16 & x < 0 \\ -2x + 3 & x \geq 0. \end{cases}$$

Take $x_n = -\frac{1}{n} \rightarrow 0$. Then

$$f(x_n) = \left(-\frac{1}{n}\right)^3 + 16 \longrightarrow 16 \neq f(0) = 3,$$

so f is discontinuous at 0. We will see below exactly what kind of discontinuity this is.

10.4 One-Sided Limits

For Example 10.4, the failure of continuity comes from the two “branches” of f disagreeing at 0 — approaching from the left gives a different answer than approaching from the right. To capture this we restrict which sequences $x_n \rightarrow a$ we are allowed to test with.

Definition 10.2.

Let $f: D \rightarrow \mathbb{R}$ and $a \in \mathbb{R}$.

- $\lim_{x \rightarrow a^-} f(x) = L$ (the **left-sided limit**) if $f(x_n) \rightarrow L$ for every *increasing* sequence $x_n \nearrow a$ with $x_n \in D$.
- $\lim_{x \rightarrow a^+} f(x) = L$ (the **right-sided limit**) if $f(x_n) \rightarrow L$ for every *decreasing* sequence $x_n \searrow a$ with $x_n \in D$.

Remark 10.2. The left- and right-sided limits may disagree. For f as in Example 10.4,

$$\lim_{x \rightarrow 0^-} f(x) = 16, \quad \lim_{x \rightarrow 0^+} f(x) = 3,$$

since $x^3 + 16 \rightarrow 16$ along any $x_n \nearrow 0$, while $-2x + 3 \rightarrow 3$ along any $x_n \searrow 0$.

Definition 10.3.

f has a **jump discontinuity** at a if both one-sided limits exist and are finite, but

$$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x).$$

This makes precise the informal description from Remark 6.3, and excludes the unbounded case of Remark 6.4, where no finite one-sided limit exists at all. The function of Example 10.4 has a jump discontinuity at 0.

Theorem 10.1.

Let $f: D \rightarrow \mathbb{R}$ and $a \in D$. Then f is continuous at a if and only if

$$\lim_{x \rightarrow a^-} f(x) = f(a) = \lim_{x \rightarrow a^+} f(x).$$

Like the two-sided limit (Proposition 10.1), each one-sided limit has an equivalent ε - δ form obtained by restricting to one side:

$$\lim_{x \rightarrow a^-} f(x) = L \quad \Leftrightarrow \quad \forall \varepsilon > 0 \exists \delta > 0 : a - \delta < x < a \quad \Rightarrow \quad |f(x) - L| < \varepsilon,$$

and symmetrically for $\lim_{x \rightarrow a^+} f(x) = L$ with $a < x < a + \delta$.

Proof. If f is continuous at a , then $f(x_n) \rightarrow f(a)$ for *every* sequence $x_n \rightarrow a$ (Proposition 10.2) — in particular for every increasing sequence $x_n \nearrow a$ and every decreasing sequence $x_n \searrow a$, giving both one-sided limits equal to $f(a)$.

Conversely, suppose both one-sided limits equal $f(a)$, and let $\varepsilon > 0$. By the ε - δ forms above there exist $\delta_1, \delta_2 > 0$ with

$$a - \delta_1 < x < a \Rightarrow |f(x) - f(a)| < \varepsilon, \quad a < x < a + \delta_2 \Rightarrow |f(x) - f(a)| < \varepsilon.$$

Let $\delta = \min(\delta_1, \delta_2)$. If $0 < |x - a| < \delta$, then either $a - \delta < x < a$ or $a < x < a + \delta$, so in both cases $|f(x) - f(a)| < \varepsilon$; and $x = a$ trivially gives $|f(x) - f(a)| = 0 < \varepsilon$. So $|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$, which is exactly Definition 6.2. $\square$

10.5 Limit Laws for Functions

Because Definition 10.1 reduces everything to sequences, the arithmetic of function limits carries over directly from the arithmetic of sequence limits (Proposition 8.1).

Proposition 10.3 (Arithmetic of function limits). *Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$. Then*

$$\lim_{x \rightarrow a} (f(x) + g(x)) = L + M, \quad \lim_{x \rightarrow a} f(x)g(x) = LM,$$

and, if $M \neq 0$,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Corollary 10.1. *If $a \in \text{dom}(f) \cap \text{dom}(g)$ and f, g are continuous at a , then $f + g$, $f \cdot g$, and (provided $g(a) \neq 0$) f/g are continuous at a .*

Example 10.5.

$$\lim_{x \rightarrow 0} \frac{x^3 - 3x + 4}{x^2 + \cos(x)} = \frac{\lim_{x \rightarrow 0} x^3 - \lim_{x \rightarrow 0} 3x + \lim_{x \rightarrow 0} 4}{\lim_{x \rightarrow 0} x^2 + \lim_{x \rightarrow 0} \cos(x)} = \frac{0 - 0 + 4}{0 + 1} = 4.$$

Example 10.6.

Careful: the quotient rule requires the *denominator's* limit to be nonzero — it says nothing when both numerator and denominator vanish. For

$$\lim_{x \rightarrow a} \frac{x^2 - a^2}{x - a},$$

we have $\lim_{x \rightarrow a} (x - a) = 0$, so Proposition 10.3 simply does not apply. Instead, factor first:

$$\frac{x^2 - a^2}{x - a} = \frac{(x - a)(x + a)}{x - a} = x + a \quad (x \neq a),$$

and only *then* take the limit: $\lim_{x \rightarrow a} (x + a) = 2a$.

10.6 Continuity of Combinations and Compositions

Proposition 10.2 lets us move freely between “ f is continuous at a ” and “ $\lim_{x \rightarrow a} f(x) = f(a)$ ”. The next fact is the one we have already been using informally whenever we wrote something like $\lim_{x \rightarrow a} \cos(h(x)) = \cos(\lim_{x \rightarrow a} h(x))$: a limit may be passed from the *outside* of a continuous function to its *inside*.

Proposition 10.4 (Limits pass through continuous functions). *Suppose $\lim_{x \rightarrow a} f(x) = L$, and g is continuous at L . Then*

$$\lim_{x \rightarrow a} g(f(x)) = g(L) = g\left(\lim_{x \rightarrow a} f(x)\right).$$

Proof. Let (x_n) be an arbitrary sequence with $x_n \in \text{dom}(f)$, $x_n \neq a$, $x_n \rightarrow a$. By hypothesis (Definition 10.1), $f(x_n) \rightarrow L$. Since g is continuous at L , feeding the sequence $(f(x_n)) \rightarrow L$ into Definition 6.2 gives $g(f(x_n)) \rightarrow g(L)$. As (x_n) was an arbitrary sequence approaching a , Definition 10.1 gives $\lim_{x \rightarrow a} g(f(x)) = g(L)$. $\square$

Taking $L = f(a)$ recovers continuity of a composition:

Corollary 10.2 (Continuity of compositions). *If f is continuous at a and g is continuous at $f(a)$, then $g \circ f$ is continuous at a .*

Proof. Since f is continuous at a , Proposition 10.2 gives $\lim_{x \rightarrow a} f(x) = f(a)$. Applying Proposition 10.4 with $L = f(a)$,

$$\lim_{x \rightarrow a} (g \circ f)(x) = \lim_{x \rightarrow a} g(f(x)) = g(f(a)) = (g \circ f)(a),$$

so $g \circ f$ is continuous at a by Proposition 10.2. $\square$

Together with Corollary 10.1 — continuity of $f + g$, $f \cdot g$, and f/g (where $g \neq 0$) from continuity of f and g — Corollary 10.2 means that *any* function built out of continuous building blocks by finitely many arithmetic operations and compositions is again continuous wherever it is defined. This is exactly the justification behind phrases like “combination of continuous functions” or “composition of continuous functions” used to evaluate a limit by plugging in, as in Example 10.5 above.

Remark 10.3. The **elementary functions** — polynomials, rational functions, power and algebraic functions, exponentials, logarithms, and the trigonometric functions — are continuous at every point of their domain. Each basic type can be checked directly from Definition 6.2, the way we checked x^3 in Example 10.2; Corollary 10.1 and Corollary 10.2 then extend continuity to every function assembled from these by arithmetic and composition, such as $\sqrt{2 + \sqrt{2x}}$ or $\cos(\pi \sin(x)/x)$. Once we

develop differentiation, this will also follow a second way: every elementary function is differentiable wherever it is defined, and differentiability will turn out to imply continuity.

10.7 The Intermediate Value Theorem

Theorem 10.2 (Intermediate Value Theorem). *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous, and suppose $a < b$ with $f(a) < 0$ and $f(b) > 0$. Then there exists $c \in (a, b)$ with $f(c) = 0$.*

Proof. We construct c by **bisection**. Set $a_1 = a$, $b_1 = b$, so $f(a_1) \leq 0 \leq f(b_1)$. Given a_n, b_n with $f(a_n) \leq 0$ and $f(b_n) \geq 0$, let $m_n = \frac{a_n + b_n}{2}$.

- If $f(m_n) = 0$, we are done — take $c = m_n$.
- If $f(m_n) < 0$, set $a_{n+1} = m_n$, $b_{n+1} = b_n$.
- If $f(m_n) > 0$, set $a_{n+1} = a_n$, $b_{n+1} = m_n$.

In either of the last two cases $f(a_{n+1}) \leq 0 \leq f(b_{n+1})$, so the invariant is preserved, and we continue. If the process never terminates, this produces sequences with (a_n) increasing, (b_n) decreasing, $a_n \leq b_n$ for all n , and

$$b_n - a_n = \frac{b - a}{2^{n-1}} \rightarrow 0$$

(the interval length halves at each step; compare Exercise 8.3). By Corollary 8.1, (a_n) and (b_n) converge to a common limit $c \in [a, b]$.

Since f is continuous, Proposition 10.2 gives $f(a_n) \rightarrow f(c)$ and $f(b_n) \rightarrow f(c)$. Now $f(a_n) \leq 0$ for every n ; if we had $f(c) > 0$, taking $\varepsilon = f(c)/2$ in the definition of convergence would force $f(a_n) > f(c)/2 > 0$ for large n , a contradiction. So $f(c) \leq 0$. The same argument with $f(b_n) \geq 0$ gives $f(c) \geq 0$. Hence $f(c) = 0$, and $a < c < b$ since $f(a) < 0$ and $f(b) > 0$ rule out $c = a$ or $c = b$. $\square$

Remark 10.4. The same bisection idea proves the more general statement: if f is continuous on $[a, b]$ and y lies between $f(a)$ and $f(b)$, then $f(c) = y$ for some $c \in [a, b]$ — apply Theorem 10.2 to $f(x) - y$.

10.8 Summary

- $\lim_{x \rightarrow a} f(x) = L$ (Definition 10.1) means $f(x_n) \rightarrow L$ for **every** sequence $x_n \rightarrow a$ in $\text{dom}(f)$ — this reduces function limits to sequence limits, and is equivalent to the ε - δ condition of Proposition 10.1.

- **Continuity restated:** f is continuous at a exactly when $\lim_{x \rightarrow a} f(x)$ exists and equals $f(a)$ (Proposition 10.2).
- **One-sided limits** (Definition 10.2) test only monotone sequences approaching from one side. A **jump discontinuity** (Definition 10.3) is exactly a point where both one-sided limits exist (finitely) but disagree; f is continuous at a iff both one-sided limits equal $f(a)$ (Theorem 10.1).
- The **limit laws** (Proposition 10.3) compute limits of sums, products, and quotients termwise, and give continuity of $f + g$, $f \cdot g$, f/g from continuity of f and g (Corollary 10.1) — but a $\frac{0}{0}$ form must be simplified *before* the quotient rule applies (Example 10.6).
- **Compositions of continuous functions are continuous** (Corollary 10.2), and more generally a limit may be passed through a continuous function from the outside in (Proposition 10.4). Together with the arithmetic rule above, this shows every **elementary function** is continuous where defined (Remark 10.3).
- The **Intermediate Value Theorem** (Theorem 10.2): a continuous function that is negative at a and positive at b must vanish somewhere in between. The proof bisects $[a, b]$ into nested intervals shrinking to a point c , then uses continuity to pass $f(a_n) \leq 0 \leq f(b_n)$ to the limit.

Part IV

The Riemann Integral

11 Riemann Sums

11.1 Area of the Circle

We start with a problem from antiquity: what is the area of a circle of radius 1? The classical idea (going back to Archimedes) is to approximate the circle by figures whose area we *can* compute — regular n -gons inscribed in the circle.

Split the n -th polygon ($n \geq 3$) into n congruent triangles, each with its apex at the center of the circle, two sides of length 1, and apex angle $\frac{2\pi}{n}$. Taking one of the two unit sides as base, the height of the triangle is $h = \sin(\frac{2\pi}{n})$, so

$$\text{area of one triangle} = \frac{1}{2} \cdot 1 \cdot h = \frac{\sin(\frac{2\pi}{n})}{2},$$

$$A_n := \text{area of the } n\text{-gon} = \frac{n}{2} \sin\left(\frac{2\pi}{n}\right).$$

As n grows, the polygons hug the circle more and more tightly, and A_n should approximate the area of the circle better and better.

Formally, we can say two things about the sequence (A_n) :

- (A_n) is **increasing**.
- (A_n) is **bounded**: every inscribed n -gon is contained in the square of side length 2 circumscribing the circle, so $A_n \leq 4$ for all n .

By the Monotone Convergence Theorem (2), (A_n) converges. Its limit is π .

Remark 11.1. Two points are glossed over here. Monotonicity is not obvious from the picture, since the n -gon is not contained in the $(n+1)$ -gon; it follows from the fact that $\sin(x)/x$ is decreasing on $(0, \pi]$, writing $A_n = \pi \cdot \frac{\sin(2\pi/n)}{2\pi/n}$. The same rewriting shows that identifying the limit as π amounts to knowing $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ — a limit we will return to when we study derivatives of the trigonometric functions.

How can we generalize this idea to other areas? Triangles could work, but it is not clear how to set this up *methodically* for an arbitrary region. For regions under the graph of a function, there is a better choice: **rectangles**, whose area is as easy to compute as it gets.

11.2 Approximation of Areas by Rectangles

Whatever approximation scheme we use, we want it to have the following properties:

- It uses a geometric figure whose area is easily computable. Rectangles are as easy as it gets.
- There is a **general method** to approximate an area by rectangles.
- There is a **general method** to **refine** any approximation.

11.2.1 Areas under Curves

Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous with $f(x) \geq 0$ for all $x \in [a, b]$. We want to compute

$A :=$ area between the graph of f and the x -axis, between $x = a$ and $x = b$.

The idea is as follows.

- **Partition** $[a, b]$ into finitely many segments

$$a = x_0 < x_1 < \cdots < x_n = b.$$

The segments $[x_{i-1}, x_i]$ will form the *bases* of rectangles.

- For each segment $[x_{i-1}, x_i]$, pick a point $x_i^* \in [x_{i-1}, x_i]$. The *height* of the i -th rectangle is $f(x_i^*)$.

If R_i denotes the area of the i -th rectangle, then

$$R_i = |x_i - x_{i-1}| \cdot f(x_i^*) \quad (i = 1, \dots, n),$$

and $R_1 + \cdots + R_n$ is an approximation to A .

Definition 11.1.

A finite set of points

$$P: \quad a = x_0 < x_1 < \cdots < x_n = b$$

is called a **partition** of $[a, b]$. Points $S = \{x_1^*, \dots, x_n^*\}$ with $x_i^* \in [x_{i-1}, x_i]$ are called **sample points**, and

$$\Delta_i = |x_i - x_{i-1}| = x_i - x_{i-1} \quad (i = 1, \dots, n)$$

is the length of the i -th segment. The **Riemann sum** for f on $[a, b]$, given P and S , is

$$RS(P, S) = f(x_1^*) \Delta_1 + f(x_2^*) \Delta_2 + \cdots + f(x_n^*) \Delta_n = \sum_{i=1}^n f(x_i^*) \Delta_i.$$

The symbol $\sum$ is **summation** (or Σ -) **notation**: $\sum_{i=1}^n c_i$ stands for $c_1 + c_2 + \cdots + c_n$. Note that Definition 11.1 makes sense for *any* function f on $[a, b]$ — neither continuity nor $f \geq 0$ is needed to *form* the sum. (If f takes negative values, the corresponding terms simply count as negative area.)

Refinement. How do we make the approximation better? We *refine the partition* — add more division points so that the segments get shorter — and choose new sample points. The goal is to find a way to do this that is both *methodological* and *practical*. The natural requirement is that the length of **every** segment tends to 0. It is convenient to measure this by the length of the longest segment,

$$\|P\| := \max_{1 \leq i \leq n} \Delta_i,$$

called the **mesh** of P . (Other authors call it the *span*.)

11.3 Equipartitions and the Definition of the Integral

It turns out the approximation process is remarkably robust — several approaches are possible, and they all give the same answer. The most practical one restricts to a special kind of partition.

Equipartitions. All segments have the *same* length, so only the number of segments matters. With n segments,

$$\Delta_i = \frac{b-a}{n} =: \Delta \quad \text{for all } i, \quad x_i = a + i\Delta.$$

There are three standard ways to choose sample points:

- **left endpoints:** $x_i^* = x_{i-1}$,
- **right endpoints:** $x_i^* = x_i$,
- **midpoints:** $x_i^* = \bar{x}_i := \frac{x_{i-1} + x_i}{2}$.

This gives three sequences of Riemann sums:

$$LHS_n = \sum_{i=1}^n f(x_{i-1}) \Delta, \quad RHS_n = \sum_{i=1}^n f(x_i) \Delta, \quad MS_n = \sum_{i=1}^n f(\bar{x}_i) \Delta,$$

the **left-hand sum**, the **right-hand sum**, and the **midpoint sum**.

Theorem 11.1.

If f is continuous on $[a, b]$, then all three limits

$$\lim_{n \rightarrow \infty} LHS_n, \quad \lim_{n \rightarrow \infty} RHS_n, \quad \lim_{n \rightarrow \infty} MS_n$$

exist and are equal.

Definition 11.2.

Let f be continuous on $[a, b]$. The **(Riemann) integral** of f from a to b is the common value L of the limits in Theorem 11.1:

$$\int_a^b f(x) dx = L = \lim_{n \rightarrow \infty} LHS_n = \lim_{n \rightarrow \infty} RHS_n = \lim_{n \rightarrow \infty} MS_n.$$

The notation is Leibniz's: the integral sign $\int$ is an elongated S for “sum”, and dx recalls the lengths Δ . It is only notation — dx on its own has no meaning here. Likewise, the name of the variable is irrelevant: $\int_a^b f(x) dx = \int_a^b f(t) dt$.

11.4 Why the Choice Does Not Matter (optional)

Theorem 11.1 is a special case of a much stronger statement: for a continuous function it does not matter at all *how* we partition or *where* we sample, as long as the segments get uniformly short.

Theorem 11.2 (Existence of the integral). *Let f be continuous on $[a, b]$. There is a unique number I with the following property: for **every** sequence of partitions $P_1, P_2, \dots$ of $[a, b]$ with $\|P_n\| \rightarrow 0$, and **every** choice of sample points S_n for P_n ,*

$$\lim_{n \rightarrow \infty} RS(P_n, S_n) = I.$$

Since an equipartition into n segments has mesh $\frac{b-a}{n} \rightarrow 0$, Theorem 11.1 follows at once, with $\int_a^b f(x) dx = I$. We now sketch *why* Theorem 11.2 is true. The argument is informal in two places, flagged in Remark 11.2 below.

11.4.1 Step 1: Trapping Every Riemann Sum

A continuous function on a closed interval attains a maximum and a minimum there. (This is the *Extreme Value Theorem*, which we have not proved yet.) So for a partition P we can put

$$m_i = \min\{f(x) : x_{i-1} \leq x \leq x_i\}, \quad M_i = \max\{f(x) : x_{i-1} \leq x \leq x_i\},$$

and form the **lower** and **upper (Darboux) sums**

$$L(P) = \sum_{i=1}^n m_i \Delta_i, \quad U(P) = \sum_{i=1}^n M_i \Delta_i.$$

Geometrically (for $f \geq 0$), $L(P)$ is the area of the largest rectangles on the segments that fit *under* the graph, and $U(P)$ is the area of the smallest rectangles that reach *over* it.

Whatever sample points x_i^* we choose, $m_i \leq f(x_i^*) \leq M_i$, so

$$L(P) \leq RS(P, S) \leq U(P) \quad \text{for every choice of sample points } S.$$

In other words, $L(P)$ and $U(P)$ are the *worst cases* over all choices of sample points. If we can show that they are close to each other, the choice of sample points becomes irrelevant.

11.4.2 Step 2: Refining Moves the Sums Toward Each Other

Suppose P' is **finer** than P , i.e. P' contains all points of P and possibly more. Adding a single point z inside a segment $[x_{i-1}, x_i]$ splits it into $[x_{i-1}, z]$ and $[z, x_i]$. The minimum of f over each smaller piece is *at least* the minimum m_i over the whole segment, and the maximum over each piece is *at most* M_i . Adding points one at a time, we get

$$L(P) \leq L(P') \quad \text{and} \quad U(P') \leq U(P).$$

So refining can only push the lower sum up and the upper sum down.

This has a useful consequence. Given *any* two partitions P and Q , the partition $P \cup Q$ is finer than both, so

$$L(P) \leq L(P \cup Q) \leq U(P \cup Q) \leq U(Q).$$

That is: **every lower sum is at most every upper sum**, no matter which partitions they come from. All lower sums sit to the left of all upper sums on the number line, and by the completeness of $\mathbb{R}$ (compare the Nested Interval Axiom) there is at least one number I in between:

$$L(P) \leq I \leq U(Q) \quad \text{for all partitions } P, Q.$$

11.4.3 Step 3: Closing the Gap

The gap between upper and lower sum is

$$U(P) - L(P) = \sum_{i=1}^n (M_i - m_i) \Delta_i.$$

Here continuity enters. On a very short segment a continuous function cannot vary much. Suppose we know the following: for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|x - y| < \delta \quad \Rightarrow \quad |f(x) - f(y)| < \varepsilon \quad \text{for all } x, y \in [a, b]. \quad (11.1)$$

Then, as soon as $\|P\| < \delta$, any two points in the same segment are within δ of each other, so $M_i - m_i < \varepsilon$ for every i , and

$$U(P) - L(P) < \sum_{i=1}^n \varepsilon \Delta_i = \varepsilon (b - a).$$

Since ε was arbitrary, the gap can be made as small as we like by making the mesh small.

11.4.4 Step 4: Conclusion

Now let $\|P_n\| \rightarrow 0$ and let S_n be any sample points. By Steps 1 and 2, both $RS(P_n, S_n)$ and I lie in the interval $[L(P_n), U(P_n)]$, so

$$|RS(P_n, S_n) - I| \leq U(P_n) - L(P_n).$$

By Step 3 the right-hand side tends to 0, so $RS(P_n, S_n) \rightarrow I$ (by the comparison test, Proposition 8.2). The number I is unique: any other I' trapped between all lower and all upper sums satisfies $|I - I'| \leq U(P_n) - L(P_n) \rightarrow 0$ as well. Nothing in this argument depended on *which* partitions or sample points we used — only on the mesh going to 0.

Remark 11.2. The sketch uses three facts we have not yet established:

1. the **Extreme Value Theorem** (continuous functions attain a max and min on closed intervals), used to define m_i and M_i ;
2. the existence of a number I between all lower and all upper sums, which is a consequence of the completeness of $\mathbb{R}$;
3. condition Equation 11.1, called **uniform continuity**.

The third point is the subtle one. It looks like Definition 6.2, but the quantifiers are in a different order: in Definition 6.2, δ is chosen *after* the point a , so it may depend on a ; in Equation 11.1, one δ must work *simultaneously for all points* of $[a, b]$. For instance, $f(x) = \frac{1}{x}$ is continuous on $(0, 1]$, but near 0 it becomes steeper and steeper, and no single δ works for a given ε . On a **closed, bounded** interval this cannot happen: a continuous $f: [a, b] \rightarrow \mathbb{R}$ is automatically uniformly continuous. Like the Extreme Value Theorem, this can be proved with a bisection / nested interval argument in the spirit of the proof of Theorem 10.2.

Remark 11.3. Continuity really matters. For the Dirichlet function of Example 10.3 on $[0, 1]$, every segment of every partition contains both rational and irrational points. Choosing all sample points rational gives $RS(P, S) = \sum_i 1 \cdot \Delta_i = 1$; choosing them all irrational gives $RS(P, S) = 0$ — for *every* partition, however fine. So the Riemann sums have no common limit, and the Dirichlet function is not Riemann integrable.

11.5 Verifying the Riemann Integral

Before computing anything new, let us check that Definition 11.2 gives the right answer for areas we already know from geometry. By Theorem 11.1 we may use whichever of the three sums is most convenient. Throughout, $\Delta = \frac{b-a}{n}$ and $x_i = a + i\Delta$.

Example 11.1.

Let $f(x) = c$. Every sample point gives height c , so

$$LHS_n = RHS_n = \sum_{i=1}^n \frac{b-a}{n} \cdot c = n \cdot \frac{b-a}{n} \cdot c = (b-a)c,$$

and hence

$$\int_a^b c \, dx = \lim_{n \rightarrow \infty} LHS_n = \lim_{n \rightarrow \infty} (b-a)c = (b-a)c,$$

the area of a rectangle with base $b-a$ and height c .

Example 11.2.

Let $f(x) = x$. Using right endpoints $x_i = a + \frac{b-a}{n}i$,

$$\begin{aligned} RHS_n &= \sum_{i=1}^n \frac{b-a}{n} \cdot f\left(a + \frac{b-a}{n}i\right) = \frac{b-a}{n} \sum_{i=1}^n \left(a + \frac{b-a}{n}i\right) \\ &= \frac{b-a}{n} \left[\left(a + \frac{b-a}{n} \cdot 1\right) + \left(a + \frac{b-a}{n} \cdot 2\right) + \cdots + \left(a + \frac{b-a}{n} \cdot n\right) \right] \\ &= \frac{b-a}{n} \left(na + \frac{b-a}{n} \cdot \frac{n(n+1)}{2} \right) \\ &= (b-a)a + \frac{b-a}{n} \cdot \frac{b-a}{n} \cdot \frac{n(n+1)}{2} \\ &= (b-a)a + \frac{(b-a)^2}{2} \cdot \frac{n+1}{n} \xrightarrow{n \rightarrow \infty} (b-a)a + \frac{(b-a)^2}{2}. \end{aligned}$$

Here we used $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ and $\frac{n+1}{n} \rightarrow 1$.

For $0 \leq a < b$ this is exactly what geometry predicts: the region under $y = x$ between a and b is a rectangle of base $b-a$ and height a (area $(b-a)a$), topped by a right triangle with both legs of length $b-a$ (area $\frac{(b-a)^2}{2}$). Simplifying,

$$\int_a^b x \, dx = (b-a) \left(a + \frac{b-a}{2}\right) = \frac{(b-a)(b+a)}{2} = \frac{b^2 - a^2}{2}.$$

Exercise 11.1. Verify this for an arbitrary linear function $f(x) = mx + c$: compute $\lim_{n \rightarrow \infty} RHS_n$ and check that the result agrees with the area of the trapezoid under the graph (when $f \geq 0$ on $[a, b]$).

11.6 “New” Functions: x^2

The examples above only reproduced areas we already knew. The first genuinely new area is the one under a parabola. Let $f(x) = x^2$. Then

$$RHS_n = \frac{b-a}{n} \sum_{i=1}^n f\left(a + \frac{b-a}{n} i\right) = \frac{b-a}{n} \sum_{i=1}^n \left(a + \frac{b-a}{n} i\right)^2.$$

To make life a little easier, take $a = 0$:

$$\begin{aligned} RHS_n &= \frac{b}{n} \sum_{i=1}^n \left(\frac{b}{n} i\right)^2 = \frac{b}{n} \cdot \frac{b^2}{n^2} (1^2 + 2^2 + \dots + n^2) \\ &= \frac{b^3}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{b^3}{6} \left(1 + \frac{1}{n}\right) \left(2 + \frac{1}{n}\right) \xrightarrow{n \rightarrow \infty} \frac{b^3 \cdot 2}{6} = \frac{b^3}{3}. \end{aligned}$$

So

$$\int_0^b x^2 dx = \frac{b^3}{3}.$$

Exercise 11.2. Prove by induction that $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \in \mathbb{N}$.

Exercise 11.3. Generalize the computation above to show that for arbitrary $a < b$,

$$\int_a^b x^2 dx = \frac{b^3 - a^3}{3}.$$

(Expand $\left(a + \frac{b-a}{n} i\right)^2$ and use the formulas for $\sum i$ and $\sum i^2$.)

11.7 Summary

- The area of the unit circle is the limit of the areas $A_n = \frac{n}{2} \sin\left(\frac{2\pi}{n}\right)$ of inscribed regular n -gons; the sequence is increasing and bounded, so it converges by 2.
- For areas under a graph we use **rectangles**: a **partition** P of $[a, b]$ and **sample points** S give the **Riemann sum** $RS(P, S) = \sum_{i=1}^n f(x_i^*) \Delta_i$ (Definition 11.1). We improve the approximation by refining the partition so that the **mesh** $\|P\| = \max_i \Delta_i$ tends to 0.

- Restricting to **equipartitions** with left endpoints, right endpoints, or midpoints gives three sequences LHS_n , RHS_n , MS_n . For continuous f they have a common limit (Theorem 11.1), which *defines* $\int_a^b f(x) dx$ (Definition 11.2).
- More generally, for continuous f **every** sequence of Riemann sums with mesh $\rightarrow 0$ converges to the same number (Theorem 11.2). The idea: every Riemann sum is trapped between the **lower and upper sums** $L(P) \leq RS(P, S) \leq U(P)$; refining pushes these toward each other; and **uniform continuity** makes the gap $U(P) - L(P)$ smaller than $\varepsilon(b - a)$ once the mesh is small (Remark 11.2). Without continuity this can fail, as the Dirichlet function shows (Remark 11.3).
- The definition reproduces known areas — $\int_a^b c dx = (b - a)c$ (Example 11.1) and $\int_a^b x dx = \frac{b^2 - a^2}{2}$ (Example 11.2) — and computes new ones: $\int_0^b x^2 dx = \frac{b^3}{3}$.

12 Lab: Riemann Sums

Math 140H, Honors Calculus I

In class we approximated the area under a curve by rectangles. This notebook lets you build those rectangles, look at them, and watch the Riemann sums approach the integral as the partition gets finer.

How to use this notebook - Cells marked IN CLASS are the core we will do together. - Cells marked HOMEWORK extend the same ideas further. - Cells marked PREDICT ask you to write down a guess before running the next code cell. Do not skip these; the point is to notice when your intuition is wrong.

START BY RUNNING THE NEXT CELL, this will load the necessary libraries and define all Python functions needed.

```
#
#
# ALWAYS RUN THIS CELL FIRST!!
#
#

import numpy as np
import sympy as sp
import matplotlib.pyplot as plt
from matplotlib.patches import Patch
from IPython.display import display, Math

plt.rcParams['figure.figsize'] = (8, 4.5)
x = sp.Symbol('x')

_METHODS = {
    'left': 'left', 'l': 'left', 'lefthand': 'left', 'left-hand': 'left',
    'right': 'right', 'r': 'right', 'righthand': 'right', 'right-hand': 'right',
    'midpoint': 'midpoint', 'mid': 'midpoint', 'm': 'midpoint', 'middle': 'midpoint',
}
```

```

def _to_exact(v):
    """Turn a number into an exact sympy number (0.25 -> 1/4, pi stays pi)."""
    v = sp.sympify(v)
    if isinstance(v, sp.Float):
        v = sp.nsimplify(v, rational=True)
    return v

def _fmt(v, digits=6):
    """Format a float compactly, e.g. 0.0625, 1.41421, -3.2e-05."""
    s = f'{float(v):.{digits}g}'
    if 'e' in s:
        mant, ex = s.split('e')
        s = rf'{mant} \cdot 10^{{{int(ex)}}}'
    return s

def _join_terms(terms, n, max_terms):
    """Join LaTeX terms with + and - signs; if n is large, abbreviate with + ... +."""
    if n > max_terms:
        terms = terms[:3] + [r'\cdots'] + terms[-2:]
    out = terms[0]
    for t in terms[1:]:
        if t.startswith('-'):
            out += ' - ' + t[1:].lstrip()
        else:
            out += ' + ' + t
    return out

def riemann_sum(f, a, b, n, method='left', x=None, show=True,
               show_integral=False, max_terms=6):
    """
    Compute (and by default draw) a Riemann sum for f on [a, b].

    Parameters
    -----
    f : sympy expression in one variable, e.g. x**2 or sp.sin(x)
    a, b : the integration bounds, a < b (numbers or sympy numbers like sp.pi)
    n : number of subintervals of equal length
    method : 'left', 'right', or 'midpoint'
    x : the variable of f (only needed if f has more than one symbol)

```

```

show : if False, just return the value (no picture, no computation lines)
show_integral : also display the exact integral and the error of the sum
max_terms : largest n for which every term of the sum is written out;
              for larger n the sum is abbreviated with "+ ... +"

Returns
-----
The value of the Riemann sum as a float.
"""
# ---- check and prepare the input -----
method = _METHODS.get(str(method).lower().replace(' ', ''))
if method is None:
    raise ValueError("method must be 'left', 'right', or 'midpoint'")
n = int(n)
if n < 1:
    raise ValueError('n must be a positive integer')
f = sp.sympify(f)
if x is None:
    syms = sorted(f.free_symbols, key=str)
    if len(syms) > 1:
        raise ValueError(f'f has several symbols {syms}; pass the variable as x=...')
    x = syms[0] if syms else sp.Symbol('x')
a, b = _to_exact(a), _to_exact(b)
if not (sp.N(a) < sp.N(b)):
    raise ValueError('need a < b')

# ---- the partition and the sample points -----
delta = (b - a) / n # exact length of each subinterval
shift = {'left': 0, 'right': 1, 'midpoint': sp.Rational(1, 2)}[method]
offset = {'left': 1, 'right': 0, 'midpoint': sp.Rational(1, 2)}[method]
# sample point x_i^* = a + (i - offset) * delta, for i = 1, ..., n

f_num = sp.lambdify(x, f, 'numpy')
a_f, b_f, d_f = float(a), float(b), float(delta)
left_edges = a_f + d_f * np.arange(n)
samples = left_edges + float(shift) * d_f
with np.errstate(all='ignore'):
    heights = np.broadcast_to(np.asarray(f_num(samples), dtype=float), samples.shape)
if not np.all(np.isfinite(heights)):
    bad = samples[~np.isfinite(heights)][0]
    raise ValueError(f'f is not defined (or infinite) at the sample point {bad:g}; '
                    'try a different method or interval')

```

```

value = float(np.sum(heights) * d_f)
if abs(value) < 1e-12 * max(1.0, float(np.sum(np.abs(heights)) * d_f)):
    value = 0.0          # the exact answer is 0; the rest is rounding noise

if not show:
    return value

# ---- the picture -----
fig, ax = plt.subplots()
colors = np.where(heights >= 0, 'tab:blue', 'tab:red')
edge = 'black' if n <= 100 else 'none'
ax.bar(left_edges, heights, width=d_f, align='edge', color=colors,
        alpha=0.35, edgecolor=edge, linewidth=0.8)
xs = np.linspace(a_f, b_f, 1000)
with np.errstate(all='ignore'):
    ys = np.array(np.broadcast_to(np.asarray(f_num(xs), dtype=float), xs.shape))
ys[~np.isfinite(ys)] = np.nan
ax.plot(xs, ys, color='black', linewidth=2, label=f'$f(\{x\}) = \{sp.latex(f)\}$')
if n <= 60:
    ax.plot(samples, heights, 'o', color='darkorange', markersize=5,
            label=r'$\{x_i\}^*, f(\{x_i\}^*)$')
ax.axhline(0, color='gray', linewidth=0.8)
pad = 0.05 * (b_f - a_f)
ax.set_xlim(a_f - pad, b_f + pad)
# if f blows up somewhere, keep the rectangles in view instead of the spike
h_lo, h_hi = min(0.0, heights.min()), max(0.0, heights.max())
span = h_hi - h_lo
y_lo, y_hi = np.nanmin(ys), np.nanmax(ys)
if span > 0 and (y_hi > h_hi + 2 * span or y_lo < h_lo - 2 * span):
    lo, hi = max(y_lo, h_lo - 0.5 * span), min(y_hi, h_hi + 0.5 * span)
    ax.set_ylim(min(lo, h_lo) - 0.05 * span, max(hi, h_hi) + 0.05 * span)
ax.set_xlabel(f'$\{x\}$')
ax.set_title(f'\{method.capitalize()\} Riemann sum with n = \{n\}')
handles, labels = ax.get_legend_handles_labels()
if np.any(heights < 0):
    handles += [Patch(color='tab:blue', alpha=0.35), Patch(color='tab:red', alpha=0.35)]
    labels += [r'$f(\{x_i\}^*) \ge 0$', r'$f(\{x_i\}^*) < 0$']
ax.legend(handles, labels, loc='best')
ax.grid(True, alpha=0.3)
plt.show()

# ---- the computation, line by line -----

```

```

A, B, D = sp.latex(a), sp.latex(b), sp.latex(delta)
fx = sp.latex(f)
sample_rule = {
    'left':      r'x_i^* = x_{i-1} = a + (i-1)\Delta',
    'right':     r'x_i^* = x_i = a + i\,\Delta',
    'midpoint':  r'x_i^* = \frac{x_{i-1} + x_i}{2} = a + \left(i - \tfrac{1}{2}\right)\Delta'
}[method]

display(Math(rf'f({x}) = {fx}, \quad [a, b] = [{A}, {B}], \quad n = {n}'))
a_paren = rf'\left({A}\right)' if sp.N(a) < 0 else A
display(Math(rf'\Delta = \frac{{b-a}}{{n}} = \frac{{B} - {a_paren}}{{n}} = {D}'
    + (rf' \approx {_fmt(d_f)}' if not delta.is_Rational else '')
    + rf', \quad x_i = a + i\,\Delta \quad (i = 0, 1, \ldots, {n})'))
display(Math(rf'\text{{{method}} sample points: }} {sample_rule}'))

# which indices get written out
idx = list(range(1, n + 1)) if n <= max_terms else [1, 2, 3, n - 1, n]
pts = [a + (i - offset) * delta for i in idx]
pts = [sp.nsimplify(p) if p.is_Rational else sp.simplify(p) for p in pts]
f_terms = [rf'f!\left({sp.latex(p)}\right)' for p in pts]
vals = [sp.nsimplify(sp.simplify(f.subs(x, p))) for p in pts]
if all(v.is_Rational and len(sp.latex(v)) <= 14 for v in vals):
    val_terms = [sp.latex(v) for v in vals]
else:
    val_terms = [_fmt(sp.N(v)) for v in vals]

general = rf'RS(P, S) = \sum_{{i=1}}^{{n}} f(x_i^*)\,\Delta'
display(Math(general + rf' = \Delta \left[ f(x_1^*) + f(x_2^*) + \cdots + f(x_{{n}}^*) \right])
display(Math(rf'\phantom{{RS(P, S)}} = {D} \left[ {_join_terms(f_terms, n, max_terms)} \right])
display(Math(rf'\phantom{{RS(P, S)}} = {D} \left[ {_join_terms(val_terms, n, max_terms)} \right])

# exact value when it is a short rational number, else a decimal
exact = None
if n <= 200:
    total = sp.nsimplify(sum(f.subs(x, a + (i - offset) * delta) for i in range(1, n + 1)))
    if total.is_Rational and len(sp.latex(total)) <= 20:
        exact = total
if exact is not None and exact.q != 1:
    result = rf'{sp.latex(exact)} \approx {_fmt(value, 10)}'
else:
    result = _fmt(value, 10)
display(Math(rf'\phantom{{RS(P, S)}} = \mathbf{{{result}}}))

```

```

if show_integral:
    I = sp.integrate(f, (x, a, b))
    I_num = float(sp.N(I))
    if isinstance(I, sp.Integral): # no closed form found
        display(Math(rf'\int_{{A}}^{{B}} {fx}\, d{x} \approx {_fmt(I_num, 10)}'))
    else:
        approx = '' if I.is_Rational else rf' \approx {_fmt(I_num, 10)}'
        display(Math(rf'\int_{{A}}^{{B}} {fx}\, d{x} = {sp.latex(I)}' + approx))
    err = value - I_num
    err_tex = '0' if abs(err) < 1e-12 else _fmt(err, 4)
    display(Math(rf'\text{{error: }} RS(P, S) - \int_{{A}}^{{B}} f\,d{x} \approx {err_tex}'))

return value

```

12.1 Recap: the setup from lecture

Let $f: [a, b] \rightarrow \mathbb{R}$.

- A partition P of $[a, b]$ is a choice of points $a = x_0 < x_1 < \dots < x_n = b$.
- $\Delta_i = x_i - x_{i-1}$ is the length of the i -th segment $[x_{i-1}, x_i]$.
- Sample points $S = \{x_1^*, \dots, x_n^*\}$ with $x_i^* \in [x_{i-1}, x_i]$ determine the heights $f(x_i^*)$ of the rectangles.
- The Riemann sum is

$$RS(P, S) = f(x_1^*) \Delta_1 + \dots + f(x_n^*) \Delta_n = \sum_{i=1}^n f(x_i^*) \Delta_i.$$

In this notebook we always split $[a, b]$ into n segments of equal length, so $\Delta_i = \Delta = \frac{b-a}{n}$ and $x_i = a + i\Delta$. There are three standard ways to pick the sample points:

method	sample point in $[x_{i-1}, x_i]$
'left'	$x_i^* = x_{i-1}$, the left endpoint
'right'	$x_i^* = x_i$, the right endpoint
'midpoint'	$x_i^* = \frac{x_{i-1} + x_i}{2}$, the midpoint

Run the next cell to define the function `riemann_sum`. You do not need to read its code to use it.

12.1.1 How to call `riemann_sum`

```
riemann_sum(f, a, b, n, method)
```

- `f` is an expression in `x`, for example `x**2`, `sp.sin(x)`, `sp.exp(-x**2)`, `1/x`, `sp.sqrt(1 - x**2)` (Recall: exponentiation in Python is `**`. The prefix `sp.` instructs python to use a library for symbolic computation called `sympy`.)
- `a`, `b` are the bounds with `a < b`; exact values like `sp.pi` or `sp.Rational(1, 3)` are fine
- `n` is the number of subintervals
- `method` is `'left'`, `'right'`, or `'midpoint'`

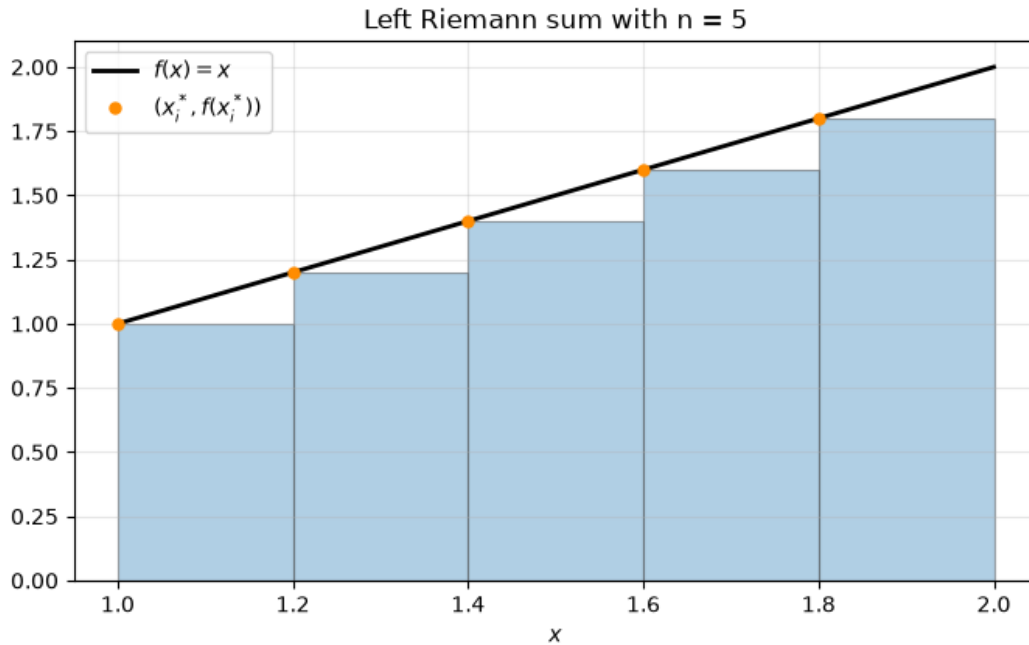
Optional arguments: - `show_integral=True` also displays the exact integral and the error of the Riemann sum - `show=False` skips the picture and just returns the number (useful in loops) - `max_terms=6` controls how many terms are written out before the sum is abbreviated with $+ \dots +$

The function returns the value of the sum. End the line with a semicolon `;` so Jupyter does not print that number a second time.

12.2 Part 1: Start

We start with $f(x) = x$ on $[1, 2]$ and $n = 5$ subintervals.

```
riemann_sum(x, 1, 2, 5, 'left');
```



$$f(x) = x, \quad [a, b] = [1, 2], \quad n = 5$$

$$\Delta = \frac{b-a}{n} = \frac{2-1}{5} = \frac{1}{5}, \quad x_i = a + i \Delta \quad (i = 0, 1, \dots, 5)$$

$$\text{left sample points: } x_i^* = x_{i-1} = a + (i-1)\Delta$$

$$\begin{aligned} RS(P, S) &= \sum_{i=1}^5 f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \dots + f(x_5^*)] \\ &= \frac{1}{5} \left[f(1) + f\left(\frac{6}{5}\right) + f\left(\frac{7}{5}\right) + f\left(\frac{8}{5}\right) + f\left(\frac{9}{5}\right) \right] \\ &= \frac{1}{5} \left[1 + \frac{6}{5} + \frac{7}{5} + \frac{8}{5} + \frac{9}{5} \right] \\ &= \frac{7}{5} \approx 1.4 \end{aligned}$$

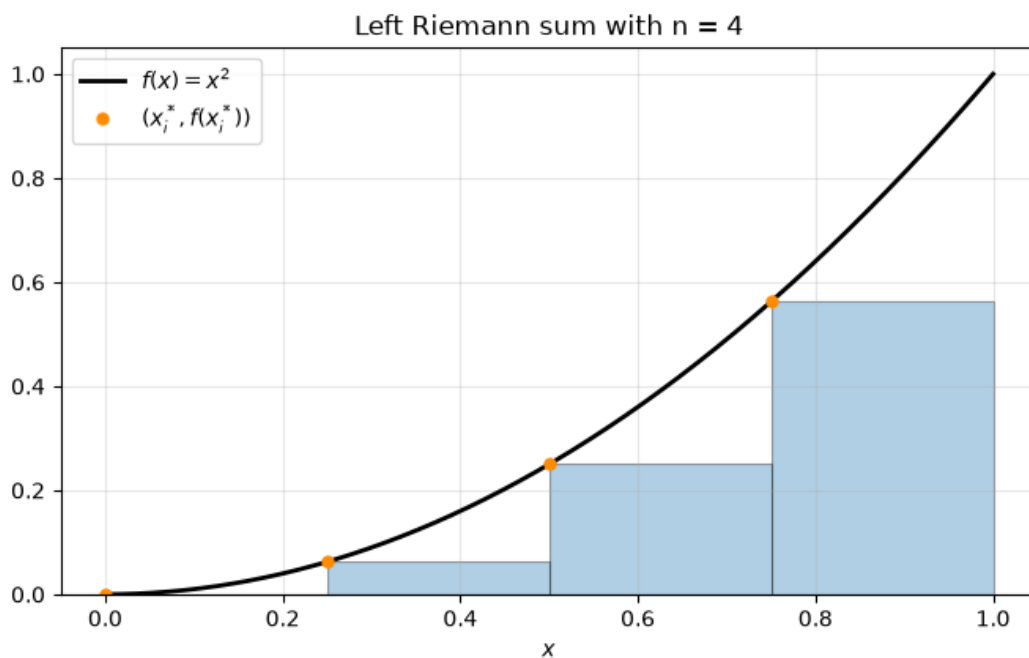
In class, we computed the actual value of the integral $\int_a^b x dx$. How does this value compare? Increase n in the cell above and see what happens.

12.3 Part 2: Left, right, and midpoint sums

Now we look at the function $f(x) = x^2$ on $[0, 1]$ and $n = 4$ subintervals.

PREDICT (on paper, before running anything): - Will the left sum overestimate or underestimate the area under x^2 ? What about the right sum? - Compute the left sum by hand. You should need nothing more than adding four fractions.

```
riemann_sum(x**2, 0, 1, 4, 'left');
```



$$f(x) = x^2, \quad [a, b] = [0, 1], \quad n = 4$$

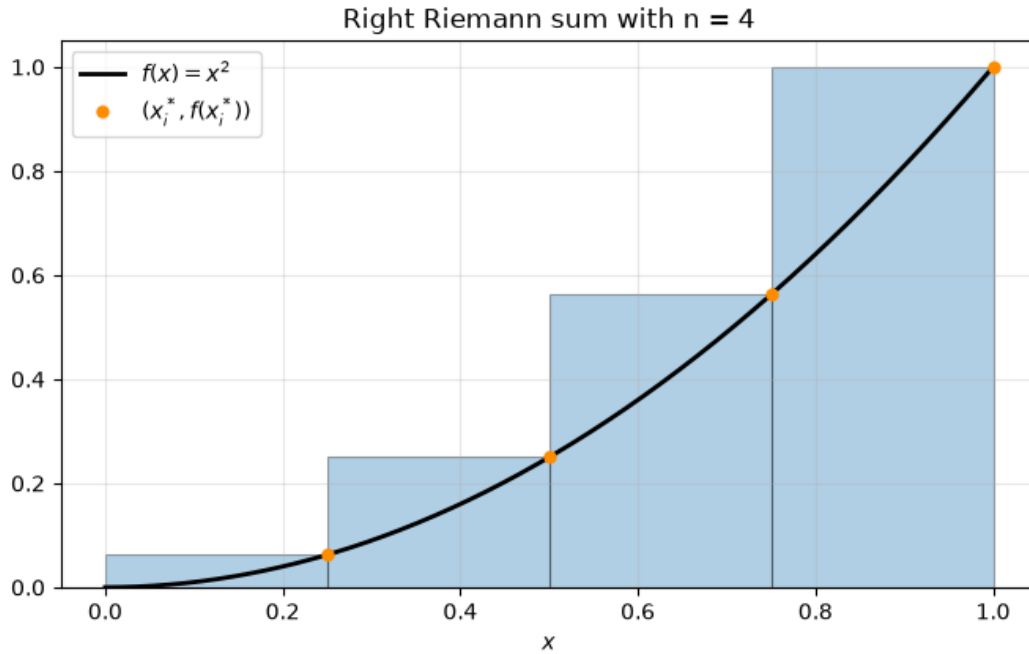
$$\Delta = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}, \quad x_i = a + i\Delta \quad (i = 0, 1, \dots, 4)$$

left sample points: $x_i^* = x_{i-1} = a + (i-1)\Delta$

$$\begin{aligned} RS(P, S) &= \sum_{i=1}^4 f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \dots + f(x_4^*)] \\ &= \frac{1}{4} \left[f(0) + f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right) \right] \\ &= \frac{1}{4} \left[0 + \frac{1}{16} + \frac{1}{4} + \frac{9}{16} \right] \end{aligned}$$

$$= \frac{7}{32} \approx 0.21875$$

```
riemann_sum(x**2, 0, 1, 4, 'right');
```



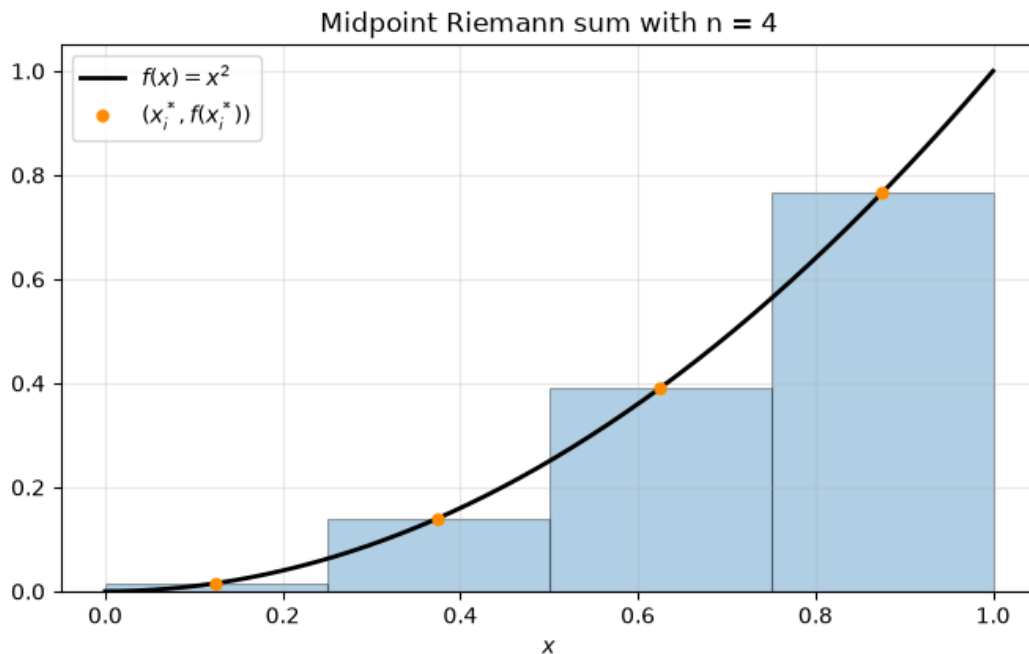
$$f(x) = x^2, \quad [a, b] = [0, 1], \quad n = 4$$

$$\Delta = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}, \quad x_i = a + i \Delta \quad (i = 0, 1, \dots, 4)$$

right sample points: $x_i^* = x_i = a + i \Delta$

$$\begin{aligned} RS(P, S) &= \sum_{i=1}^4 f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \dots + f(x_4^*)] \\ &= \frac{1}{4} \left[f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right) + f(1) \right] \\ &= \frac{1}{4} \left[\frac{1}{16} + \frac{1}{4} + \frac{9}{16} + 1 \right] \\ &= \frac{15}{32} \approx 0.46875 \end{aligned}$$

```
riemann_sum(x**2, 0, 1, 4, 'midpoint', show_integral=True);
```



$$f(x) = x^2, \quad [a, b] = [0, 1], \quad n = 4$$

$$\Delta = \frac{b-a}{n} = \frac{1-0}{4} = \frac{1}{4}, \quad x_i = a + i \Delta \quad (i = 0, 1, \dots, 4)$$

$$\text{midpoint sample points: } x_i^* = \frac{x_{i-1} + x_i}{2} = a + \left(i - \frac{1}{2}\right) \Delta$$

$$RS(P, S) = \sum_{i=1}^4 f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \dots + f(x_4^*)]$$

$$= \frac{1}{4} \left[f\left(\frac{1}{8}\right) + f\left(\frac{3}{8}\right) + f\left(\frac{5}{8}\right) + f\left(\frac{7}{8}\right) \right]$$

$$= \frac{1}{4} \left[\frac{1}{64} + \frac{9}{64} + \frac{25}{64} + \frac{49}{64} \right]$$

$$= \frac{21}{64} \approx \mathbf{0.328125}$$

$$\int_0^1 x^2 dx = \frac{1}{3}$$

$$\text{error: } RS(P, S) - \int_0^1 f dx \approx -0.005208$$

VERIFY: The left and right sums share almost all of their terms. Show that for any f and equally spaced partition,

$$RS_{\text{right}} - RS_{\text{left}} = \Delta (f(b) - f(a)),$$

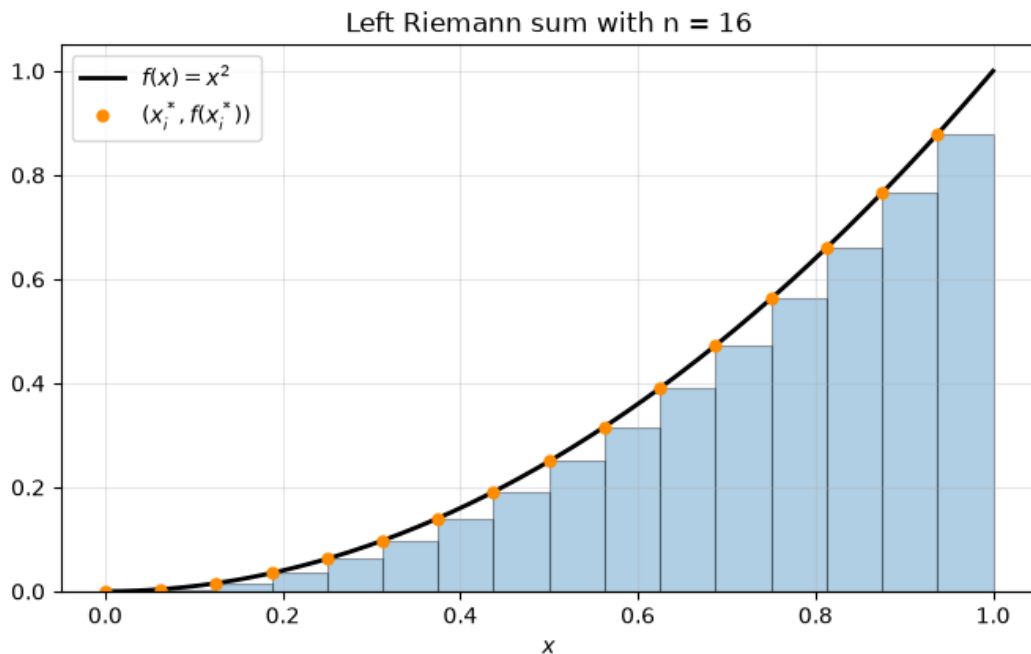
and check this against the two numbers you just computed.

REFLECT: Your prediction about over- and underestimates depended on a property of x^2 on $[0, 1]$. Which property? State a general claim about left and right sums for functions with that property.

12.4 Part 3: Refining the partition

In lecture, the approximation got better by refining the partition. Let us watch that happen. Once n is larger than 6, the written-out sum is abbreviated with $+\dots+$.

```
riemann_sum(x**2, 0, 1, 16, 'left', show_integral=True);
```



$$f(x) = x^2, \quad [a, b] = [0, 1], \quad n = 16$$

$$\Delta = \frac{b-a}{n} = \frac{1-0}{16} = \frac{1}{16}, \quad x_i = a + i\Delta \quad (i = 0, 1, \dots, 16)$$

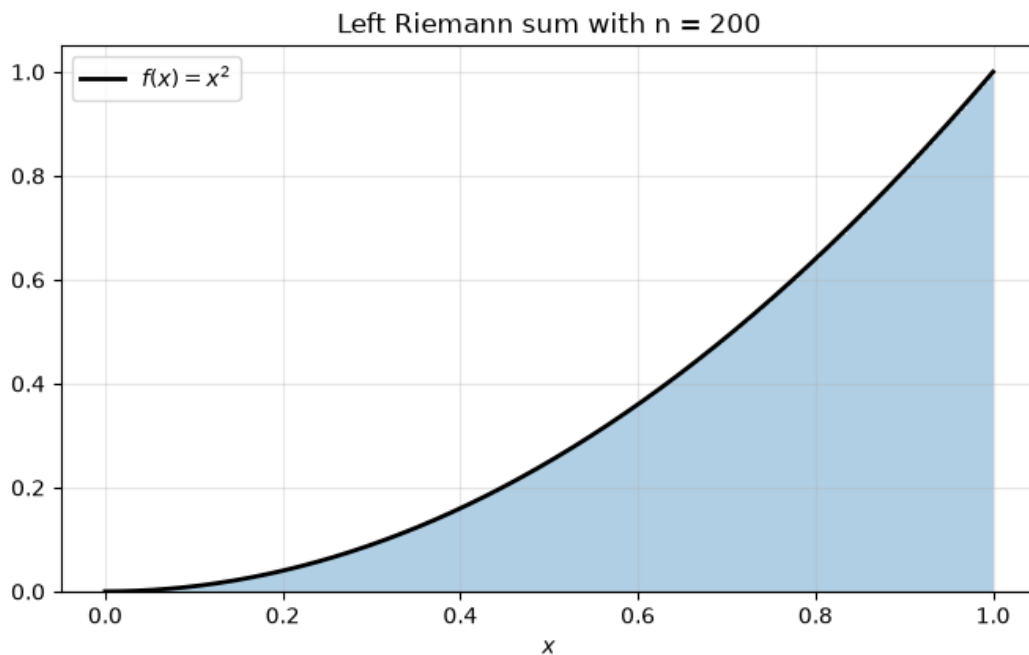
$$\text{left sample points: } x_i^* = x_{i-1} = a + (i-1)\Delta$$

$$\begin{aligned}
RS(P, S) &= \sum_{i=1}^{16} f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \cdots + f(x_{16}^*)] \\
&= \frac{1}{16} \left[f(0) + f\left(\frac{1}{16}\right) + f\left(\frac{1}{8}\right) + \cdots + f\left(\frac{7}{8}\right) + f\left(\frac{15}{16}\right) \right] \\
&= \frac{1}{16} [0 + 0.00390625 + 0.015625 + \cdots + 0.765625 + 0.878906] \\
&= \frac{155}{512} \approx \mathbf{0.302734375}
\end{aligned}$$

$$\int_0^1 x^2 dx = \frac{1}{3}$$

$$\text{error: } RS(P, S) - \int_0^1 f dx \approx -0.0306$$

```
riemann_sum(x**2, 0, 1, 200, 'left', show_integral=True);
```



$$\begin{aligned}
f(x) &= x^2, \quad [a, b] = [0, 1], \quad n = 200 \\
\Delta &= \frac{b-a}{n} = \frac{1-0}{200} = \frac{1}{200}, \quad x_i = a + i \Delta \quad (i = 0, 1, \dots, 200) \\
\text{left sample points: } x_i^* &= x_{i-1} = a + (i-1)\Delta
\end{aligned}$$

$$\begin{aligned}
RS(P, S) &= \sum_{i=1}^{200} f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \cdots + f(x_{200}^*)] \\
&= \frac{1}{200} \left[f(0) + f\left(\frac{1}{200}\right) + f\left(\frac{1}{100}\right) + \cdots + f\left(\frac{99}{100}\right) + f\left(\frac{199}{200}\right) \right] \\
&= \frac{1}{200} [0 + 2.5 \cdot 10^{-5} + 0.0001 + \cdots + 0.9801 + 0.990025] \\
&= \frac{26467}{80000} \approx \mathbf{0.3308375}
\end{aligned}$$

$$\int_0^1 x^2 dx = \frac{1}{3}$$

$$\text{error: } RS(P, S) - \int_0^1 f dx \approx -0.002496$$

12.5 Part 4: Back to the circle

We started the Riemann integral lecture by approximating the area π of the unit circle with regular n -gons. The quarter of the unit disk in the first quadrant is the region under $f(x) = \sqrt{1-x^2}$ on $[0, 1]$, so $4 \int_0^1 \sqrt{1-x^2} dx = \pi$.

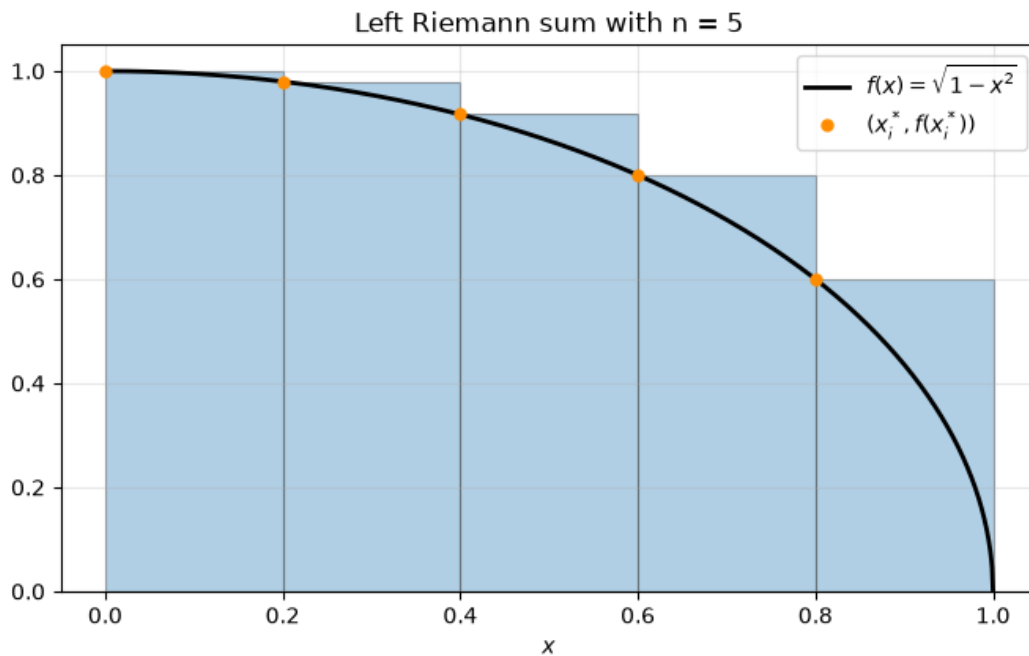
PREDICT: Is the left sum of $\sqrt{1-x^2}$ on $[0, 1]$ an overestimate or an underestimate? Why?

```

riemann_sum(sp.sqrt(1 - x**2), 0, 1, 5, 'left', show_integral=True);

# TODO: find the smallest n for which 4 * (left sum) is within 10**(-3) of pi.
#   Use riemann_sum(..., show=False) inside a loop.
# TODO: do the same for the midpoint sum. How much smaller is the n you need?

```



$$f(x) = \sqrt{1-x^2}, \quad [a, b] = [0, 1], \quad n = 5$$

$$\Delta = \frac{b-a}{n} = \frac{1-0}{5} = \frac{1}{5}, \quad x_i = a + i \Delta \quad (i = 0, 1, \dots, 5)$$

$$\text{left sample points: } x_i^* = x_{i-1} = a + (i-1)\Delta$$

$$\begin{aligned} RS(P, S) &= \sum_{i=1}^5 f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \dots + f(x_5^*)] \\ &= \frac{1}{5} \left[f(0) + f\left(\frac{1}{5}\right) + f\left(\frac{2}{5}\right) + f\left(\frac{3}{5}\right) + f\left(\frac{4}{5}\right) \right] \\ &= \frac{1}{5} [1 + 0.979796 + 0.916515 + 0.8 + 0.6] \\ &= \mathbf{0.8592622072} \end{aligned}$$

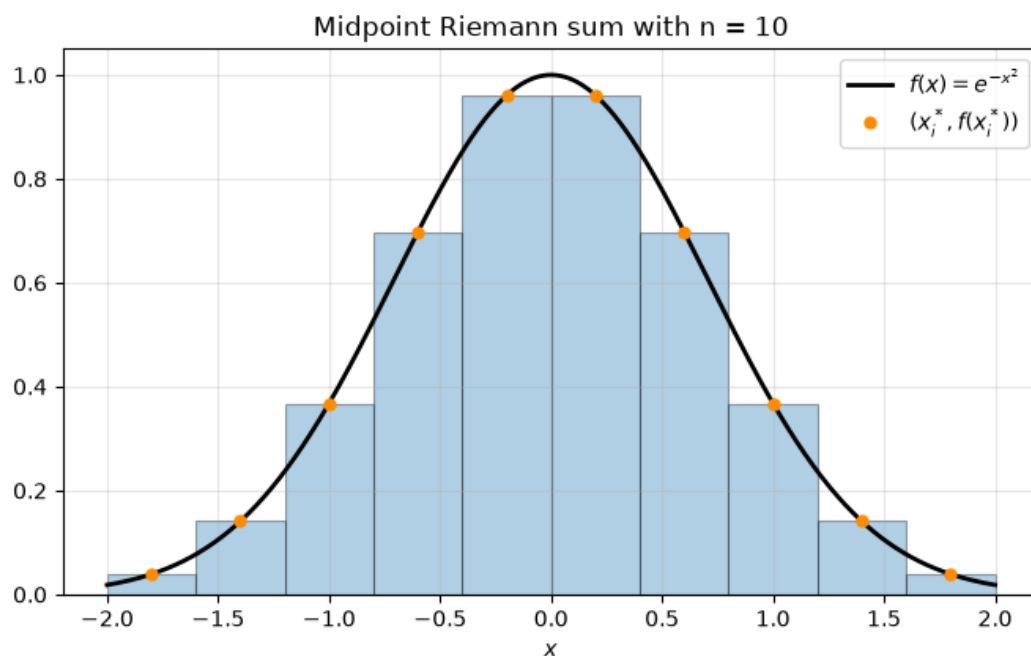
$$\int_0^1 \sqrt{1-x^2} dx = \frac{\pi}{4} \approx 0.7853981634$$

$$\text{error: } RS(P, S) - \int_0^1 f dx \approx 0.07386$$

12.6 Part 5: Experiment on your own

Change the function, bounds, n , and method in the cell below and run it again. If `ipywidgets` is installed, the cell after it gives you sliders instead.

```
f = sp.exp(-x**2)
riemann_sum(f, -2, 2, 10, 'midpoint', show_integral=True);
```



$$f(x) = e^{-x^2}, \quad [a, b] = [-2, 2], \quad n = 10$$

$$\Delta = \frac{b-a}{n} = \frac{2-(-2)}{10} = \frac{2}{5}, \quad x_i = a + i \Delta \quad (i = 0, 1, \dots, 10)$$

$$\text{midpoint sample points: } x_i^* = \frac{x_{i-1} + x_i}{2} = a + \left(i - \frac{1}{2}\right) \Delta$$

$$\begin{aligned} RS(P, S) &= \sum_{i=1}^{10} f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \dots + f(x_{10}^*)] \\ &= \frac{2}{5} \left[f\left(-\frac{9}{5}\right) + f\left(-\frac{7}{5}\right) + f(-1) + \dots + f\left(\frac{7}{5}\right) + f\left(\frac{9}{5}\right) \right] \\ &= \frac{2}{5} [0.0391639 + 0.140858 + 0.367879 + \dots + 0.140858 + 0.0391639] \\ &= \mathbf{1.765094018} \end{aligned}$$

$$\int_{-2}^2 e^{-x^2} dx = \sqrt{\pi} \operatorname{erf}(2) \approx 1.764162782$$

$$\text{error: } RS(P, S) - \int_{-2}^2 f dx \approx 0.0009312$$

```
# Interactive version: move the slider or change the method
try:
    from ipywidgets import interact, IntSlider, Dropdown

    def explore(n=4, method='left'):
        riemann_sum(sp.sin(x) + 1.5, 0, 2*sp.pi, n, method, show_integral=True)

    interact(explore,
             n=IntSlider(min=1, max=100, step=1, value=4),
             method=Dropdown(options=['left', 'right', 'midpoint'], value='left'));
except ImportError:
    print('ipywidgets is not installed; use the cell above instead.')
```

```
interactive(children=(IntSlider(value=4, description='n', min=1), Dropdown(description='method', value='left')))
```

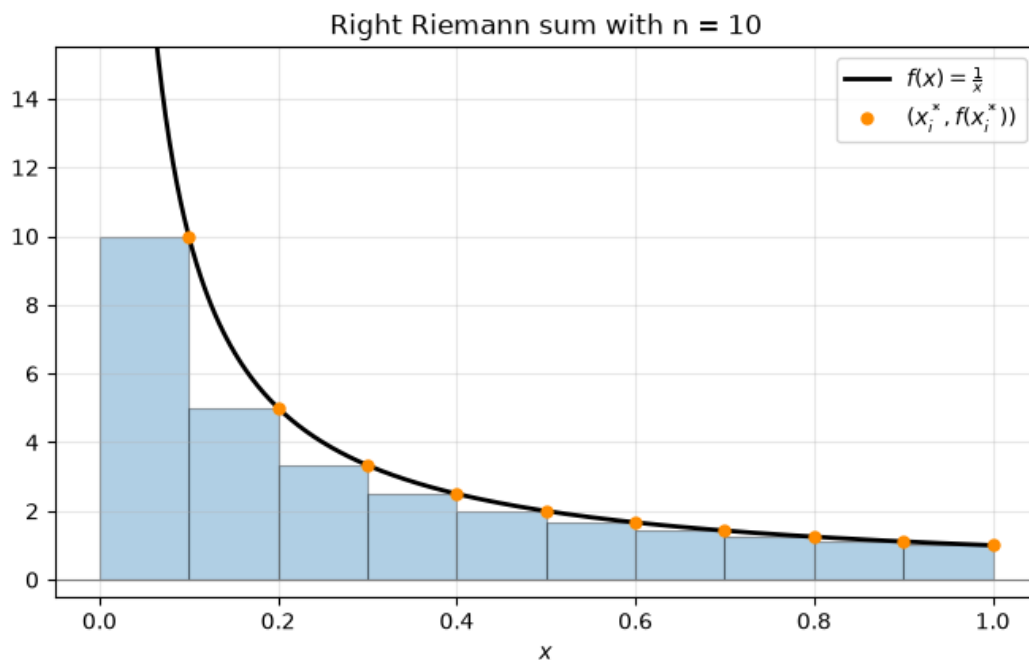
12.7 Part 6: when Riemann sums fail

Riemann sums are defined for bounded functions. Look at what happens for $f(x) = 1/x$ on $[0, 1]$ using right sums (the left sum would need $f(0)$, which does not exist).

```
riemann_sum(1/x, 0, 1, 10, 'right');

for n in [10, 100, 1000, 10000]:
    print(n, riemann_sum(1/x, 0, 1, n, 'right', show=False))

# TODO: what happens to the sums as n grows? Compare with 1/sp.sqrt(x) on [0, 1].
# The first rectangle is always very tall in both cases,
# so why do the two behave so differently?
```



$$f(x) = \frac{1}{x}, \quad [a, b] = [0, 1], \quad n = 10$$

$$\Delta = \frac{b-a}{n} = \frac{1-0}{10} = \frac{1}{10}, \quad x_i = a + i \Delta \quad (i = 0, 1, \dots, 10)$$

right sample points: $x_i^* = x_i = a + i \Delta$

$$\begin{aligned}
 RS(P, S) &= \sum_{i=1}^{10} f(x_i^*) \Delta = \Delta [f(x_1^*) + f(x_2^*) + \dots + f(x_{10}^*)] \\
 &= \frac{1}{10} \left[f\left(\frac{1}{10}\right) + f\left(\frac{1}{5}\right) + f\left(\frac{3}{10}\right) + \dots + f\left(\frac{9}{10}\right) + f(1) \right] \\
 &= \frac{1}{10} \left[10 + 5 + \frac{10}{3} + \dots + \frac{10}{9} + 1 \right] \\
 &= \frac{7381}{2520} \approx 2.928968254
 \end{aligned}$$

10 2.9289682539682538
 100 5.187377517639621
 1000 7.485470860550345
 10000 9.78760603604438

13 Properties of the Integral

Companion reading

This lecture corresponds to C&J §2.3.a–d: additivity, the integral of a sum and of a constant multiple, estimating integrals, and the mean value theorem for integrals. The table of basic integrals draws on C&J §2.2.c–e.

13.1 The Integral as Signed Area

Remark 13.1. Nothing in Definition 11.1 or Definition 11.2 requires $f \geq 0$: the integral is defined for every continuous f on $[a, b]$. But the identification

$$\int_a^b f(x) dx = \text{area under the graph of } f$$

is only valid when $f \geq 0$. If f takes negative values, the corresponding terms $f(x_i^*) \Delta_i$ of a Riemann sum are negative. So in general the integral measures **signed area**: area above the x -axis counts positively, area below the x -axis counts negatively.

Example 13.1.

Let $f(x) = x$ and $b > 0$. By Example 11.2,

$$\int_{-b}^b x dx = \frac{b^2 - (-b)^2}{2} = 0.$$

Geometrically, the region between the graph and the x -axis consists of two triangles of area $\frac{b^2}{2}$ each: one above the axis over $[0, b]$ (positive area) and one below the axis over $[-b, 0]$ (negative area). They cancel.

If we want the **total** (unsigned) area between the graph of f and the x -axis, we can compute $\int_a^b |f(x)| dx$

13.2 Additivity

Proposition 13.1 (Additivity). *Let f be continuous on $[a, b]$ and $a < c < b$. Then*

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

For $f \geq 0$ this is what the area interpretation predicts: the vertical line $x = c$ cuts the region under the graph into a part A_1 over $[a, c]$ and a part A_2 over $[c, b]$, and the total area is $A_1 + A_2$.

Proof. The idea is that any Riemann sum for $[a, b]$ in which c is a division point splits into a Riemann sum for $[a, c]$ plus one for $[c, b]$. If $c = x_m$ is a point of the partition $a = x_0 < \dots < x_n = b$, then

$$\sum_{i=1}^n f(x_i^*) \Delta_i = \sum_{i=1}^m f(x_i^*) \Delta_i + \sum_{i=m+1}^n f(x_i^*) \Delta_i,$$

where the first sum on the right is a Riemann sum for a partition of $[a, c]$ into m segments, and the second one is a Riemann sum for a partition of $[c, b]$ into $n - m$ segments.

For equipartitions of $[a, b]$, the point c is usually not a division point. But by Theorem 11.2 we may use *any* partitions whose mesh tends to 0. So let P_n consist of an equipartition of $[a, c]$ into n segments followed by an equipartition of $[c, b]$ into n segments, and let S_n be the right endpoints of its segments. Then $\|P_n\| = \max(\frac{c-a}{n}, \frac{b-c}{n}) \rightarrow 0$, and the splitting above reads

$$RS(P_n, S_n) = RHS_n(f, [a, c]) + RHS_n(f, [c, b]).$$

Letting $n \rightarrow \infty$, the left side tends to $\int_a^b f$ by Theorem 11.2, and the right side tends to $\int_a^c f + \int_c^b f$ by Definition 11.2 and the limit laws (Proposition 8.1). $\square$

Rearranging gives a formula that is often helpful for computing areas.

Corollary 13.1.

Let f be continuous on $[a, b]$ and $a < c < b$. Then

$$\int_c^b f(x) dx = \int_a^b f(x) dx - \int_a^c f(x) dx.$$

Example 13.2.

For $0 < a < b$, the computation of $\int_0^b x^2 dx = \frac{b^3}{3}$ from the previous chapter and Corollary 13.1 give

$$\int_a^b x^2 dx = \int_0^b x^2 dx - \int_0^a x^2 dx = \frac{b^3}{3} - \frac{a^3}{3},$$

without redoing the Riemann sum computation of Exercise 11.3.

13.3 Integrals with $a \geq b$

So far $\int_a^b f(x) dx$ was only defined for $a < b$. What if $a = b$ or $a > b$?

In the case $a = b$, we see that $\Delta = \frac{b-a}{n} = 0$ for any n , meaning that all Riemann sums are 0, too, which in turn makes the integral 0.

If $a > b$, Δ becomes negative. The partition points

$$a, a + \Delta, a + 2\Delta, \dots, a + n\Delta = b,$$

now count from *right to left*. And since Δ is negative, the Riemann sum will be (-1) times the sum we get when we count from left to right. This yields $\int_b^a f = -\int_a^b f$.

Definition 13.1.

Let f be continuous on an interval containing a and c .

- $\int_a^a f(x) dx = 0$ (this only requires f to be defined at a).
- If $a < b$, then $\int_b^a f(x) dx = -\int_a^b f(x) dx$.

Exercise 13.1. Let f be continuous on an interval I . Using Proposition 13.1 and Definition 13.1, show that

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

holds for **all** $a, b, c \in I$, in any order and also when some of them coincide. (Hint: there are six orderings of three distinct points. For $b < c$ with $a < b$, say, apply Proposition 13.1 to $a < b < c$.)

13.4 Sums and Constant Multiples

Proposition 13.2 (Sum rule). *Let f and g be continuous on $[a, b]$. Then*

$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx.$$

Proof. This follows directly from the definition via Riemann sums. For the right-hand sums with $\Delta = \frac{b-a}{n}$,

$$\begin{aligned} RHS_n(f + g) &= \sum_{i=1}^n (f(a + i\Delta) + g(a + i\Delta)) \Delta \\ &= \sum_{i=1}^n f(a + i\Delta) \Delta + \sum_{i=1}^n g(a + i\Delta) \Delta = RHS_n(f) + RHS_n(g). \end{aligned}$$

By the limit laws (Proposition 8.1),

$$\int_a^b (f + g) = \lim_{n \rightarrow \infty} RHS_n(f + g) = \lim_{n \rightarrow \infty} (RHS_n(f) + RHS_n(g)) \quad (13.1)$$

$$= \lim_{n \rightarrow \infty} RHS_n(f) + \lim_{n \rightarrow \infty} RHS_n(g) = \int_a^b f + \int_a^b g. \quad (13.2)$$

□

Proposition 13.3 (Constant multiple rule). *Let f be continuous on $[a, b]$ and $c \in \mathbb{R}$. Then*

$$\int_a^b c f(x) dx = c \int_a^b f(x) dx.$$

The proof is the same: $RHS_n(cf) = c \cdot RHS_n(f)$, and we let $n \rightarrow \infty$. Combining the two rules, $\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g$ for all constants α, β ; in particular $\int_a^b (f - g) = \int_a^b f - \int_a^b g$. Both rules remain true for $a \geq b$, by Definition 13.1.

Remark 13.2. What about $\int_a^b f(x) g(x) dx$? There is **no** analogous rule: the integral of a product is in general *not* the product of the integrals. For instance, with $f(x) = g(x) = x$ on $[0, 1]$,

$$\int_0^1 x \cdot x dx = \frac{1}{3} \quad \text{but} \quad \int_0^1 x dx \cdot \int_0^1 x dx = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

The trouble is already visible in the Riemann sums: $\sum f(x_i)g(x_i)\Delta$ does not factor into $(\sum f(x_i)\Delta)(\sum g(x_i)\Delta)$.

13.5 Basic Rules for Integrals

We collect the integrals of the elementary functions. Some of them can be computed directly from Riemann sums: we have done x^0 , x^1 , x^2 (Example 11.1, Example 11.2, and the x^2 computation of the previous chapter), and C&J §2.2.c–e carry out x^α and $\sin$, $\cos$. All of them will become easy consequences of the Fundamental Theorem of Calculus. For now we record them and use them freely.

Proposition 13.4 (Integrals of elementary functions).

$$\begin{aligned}\int_a^b x^\alpha dx &= \frac{1}{\alpha+1}(b^{\alpha+1} - a^{\alpha+1}) \quad (\alpha \neq -1), \\ \int_a^b x^{-1} dx &= \ln b - \ln a, \\ \int_a^b \sin x dx &= -(\cos b - \cos a), \\ \int_a^b \cos x dx &= \sin b - \sin a, \\ \int_a^b e^x dx &= e^b - e^a.\end{aligned}$$

Here the interval of integration must lie inside the domain on which the integrand is continuous: $0 < a < b$ for x^{-1} and for general real α ; any a, b if α is a nonnegative integer; a, b on the same side of 0 if α is a negative integer.

What about $\int_a^b \ln x dx$? We will come back to it later.

Combining Proposition 13.4 with the sum and constant multiple rules, we can integrate a wide range of functions, for instance every polynomial.

Example 13.3.

$$\begin{aligned}\int_a^b (2x^3 - 4 \sin x) dx &= 2 \int_a^b x^3 dx - 4 \int_a^b \sin x dx \\ &= 2 \left(\frac{1}{4}(b^4 - a^4) \right) - 4 \left(-(\cos b - \cos a) \right) \\ &= \frac{1}{2}(b^4 - a^4) + 4(\cos b - \cos a).\end{aligned}$$

Remark 13.3. This leaves open how to integrate products $f(x)g(x)$ (Remark 13.2) and compositions $f(g(x))$. Both questions will be answered later, by *integration by parts* and *substitution*.

13.6 Estimating Integrals

Throughout this section, $a < b$.

Proposition 13.5.

If $f(x) \geq 0$ for all $x \in [a, b]$, then

$$\int_a^b f(x) dx \geq 0.$$

Proof. Every term $f(x_i^*) \Delta_i$ of every Riemann sum for f is ≥ 0 , so every Riemann sum is ≥ 0 . A limit of nonnegative numbers is nonnegative (compare the argument at the end of the proof of Theorem 10.2). $\square$

Proposition 13.6.

If $f(x) \geq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx.$$

Proof. Apply Proposition 13.5 to $f - g \geq 0$ and use $\int_a^b f - \int_a^b g = \int_a^b (f - g)$ (Proposition 13.2, Proposition 13.3). $\square$

Now let $f: [a, b] \rightarrow \mathbb{R}$ be continuous. As we will show later (this is the *Extreme Value Theorem*, see Remark 11.2), f assumes a maximum and a minimum on $[a, b]$. In particular, there exist m, M such that

$$m \leq f(x) \leq M \quad \text{for all } x \in [a, b].$$

Interpret m and M as constant functions on $[a, b]$ and apply Proposition 13.6:

$$\int_a^b m dx \leq \int_a^b f(x) dx \leq \int_a^b M dx.$$

For any constant C , $\int_a^b C dx = C \int_a^b 1 dx = C(b - a)$ (Example 11.1). Hence:

Corollary 13.2 (Min–max bound). *Let f be continuous on $[a, b]$, with minimum value m and maximum value M . Then*

$$m(b - a) \leq \int_a^b f(x) dx \leq M(b - a).$$

Geometrically (for $f \geq 0$), $m(b-a)$ is the area of the rectangle of height m *inscribed* under the graph, and $M(b-a)$ is the area of the rectangle of height M *circumscribing* it. The argument only used $m \leq f \leq M$, so the bound holds for any lower bound m and upper bound M of f on $[a, b]$; the minimum and maximum give the best bound of this kind.

This bound is simple but immensely useful. Crude on a single interval, it becomes powerful when applied on many small intervals and added up.

Example 13.4 (Harmonic series).

What happens to the sequence

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

(the partial sums of the **harmonic series**) as $n \rightarrow \infty$?

Let $f(x) = \frac{1}{x}$ on $[k, k+1]$, $k \in \mathbb{N}$. Since f is decreasing, its minimum and maximum on this interval are $m = \frac{1}{k+1}$ and $M = \frac{1}{k}$, attained at the endpoints. Corollary 13.2 gives

$$\frac{1}{k+1} = \frac{1}{k+1}((k+1) - k) \leq \int_k^{k+1} \frac{1}{x} dx \leq \frac{1}{k}((k+1) - k) = \frac{1}{k}.$$

Sum all these bounds for $k = 1, \dots, n$:

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n+1} \leq \int_1^2 \frac{1}{x} dx + \int_2^3 \frac{1}{x} dx + \cdots + \int_n^{n+1} \frac{1}{x} dx \leq 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

By repeated use of Proposition 13.1, the integrals in the middle combine into a single integral, and by Proposition 13.4,

$$\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n+1} \leq \int_1^{n+1} \frac{1}{x} dx = \ln(n+1) - \ln 1 = \ln(n+1) \leq a_n.$$

Since $\ln$ is increasing with range $\mathbb{R}$ (Definition 5.5), for every C there is an N with $\ln(N+1) > C$, and then $a_n \geq \ln(n+1) > C$ for all $n \geq N$. So $a_n \rightarrow \infty$: the harmonic series diverges.

13.7 The Integral as a Mean Value

Consider the right-hand sum with $\Delta = \frac{b-a}{n}$ and $x_i = a + i\Delta$:

$$RHS_n(f) = \sum_{i=1}^n f(x_i) \Delta = \frac{b-a}{n} (f(x_1) + \cdots + f(x_n)) = (b-a) \cdot \frac{f(x_1) + \cdots + f(x_n)}{n}.$$

The fraction on the right is the **average** (or **arithmetic mean**) of the n values $f(x_1), \dots, f(x_n)$. So

$$\frac{RHS_n(f)}{b-a} = \text{avg}(f(x_1), \dots, f(x_n))$$

is the average of f over n equally spaced sample points — a “level- n average”. As $n \rightarrow \infty$, $RHS_n(f) \rightarrow \int_a^b f(x) dx$, hence

$$\frac{RHS_n(f)}{b-a} \rightarrow \frac{1}{b-a} \int_a^b f(x) dx.$$

The limit is an “ ∞ -average”: the average of *all* values of f on $[a, b]$.

Definition 13.2.

Let f be continuous on $[a, b]$. The **average** or **mean value** of f on $[a, b]$ is

$$\text{avg}(f, [a, b]) := \frac{1}{b-a} \int_a^b f(x) dx.$$

Example 13.5.

Using Proposition 13.4:

$$\text{avg}(x, [0, 1]) = \frac{1}{1} \int_0^1 x dx = \frac{1^2}{2} - \frac{0^2}{2} = \frac{1}{2},$$

$$\text{avg}(\sin x, [0, 2\pi]) = \frac{1}{2\pi} \int_0^{2\pi} \sin x dx = \frac{1}{2\pi} \left(-(\cos 2\pi - \cos 0) \right) = \frac{1}{2\pi} (- (1 - 1)) = 0,$$

$$\text{avg}(\sin x, [0, \pi]) = \frac{1}{\pi} \left(-(\cos \pi - \cos 0) \right) = \frac{1}{\pi} (-(-1 - 1)) = \frac{2}{\pi},$$

$$\text{avg}(x^2, [0, 2]) = \frac{1}{2} \int_0^2 x^2 dx = \frac{1}{2} \left(\frac{2^3}{3} - \frac{0^3}{3} \right) = \frac{4}{3}.$$

Over the full period $[0, 2\pi]$ the positive and negative areas of $\sin$ cancel (Remark 13.1), so the average is 0. Over $[0, \pi]$, where $\sin \geq 0$, the average $\frac{2}{\pi} \approx 0.64$ lies strictly between the minimum 0 and the maximum 1.

Connection with the min–max bound. Let f be continuous on $[a, b]$. Then f assumes its minimum m and its maximum M on $[a, b]$: there exist $x_m, x_M \in [a, b]$ with

$$f(x_m) = m, \quad f(x_M) = M, \quad \text{and} \quad m \leq f(x) \leq M \quad \text{for all } a \leq x \leq b.$$

Dividing the inequality of Corollary 13.2 by $b - a > 0$ gives

$$m \leq \underbrace{\frac{1}{b-a} \int_a^b f(x) dx}_{=\text{avg}(f, [a, b])} \leq M.$$

The average of a continuous function lies between its smallest and its largest value. By the Intermediate Value Theorem, f assumes **every** value between m and M — in particular its average.

Theorem 13.1 (Mean Value Theorem for Integrals). *Let f be continuous on $[a, b]$. Then there exists $c \in [a, b]$ with*

$$\frac{1}{b-a} \int_a^b f(x) dx = f(c), \quad \text{i.e.} \quad \int_a^b f(x) dx = f(c)(b-a).$$

Proof. Let $\mu = \text{avg}(f, [a, b])$. As shown above, $f(x_m) = m \leq \mu \leq M = f(x_M)$. Apply the general form of the Intermediate Value Theorem (Remark 10.4) to f on the closed interval with endpoints x_m and x_M (if $x_m = x_M$, then $m = M$, f is constant, and any c works). It yields a point c between x_m and x_M , hence in $[a, b]$, with $f(c) = \mu$. $\square$

Geometrically (for $f \geq 0$): the area under the graph of f equals the area of the rectangle over $[a, b]$ of height $f(c)$. The parts of the region under the graph that stick out above the rectangle exactly make up for the parts of the rectangle that lie above the graph.

Remark 13.4.

1. The theorem only asserts that *some* c exists; it gives no information about where it is, and c need not be unique. For $\sin$ on $[0, 2\pi]$ (Example 13.5), each of $c = 0, \pi, 2\pi$ works.
2. Continuity is essential. Let $f(x) = 0$ for $x < 1$ and $f(x) = 1$ for $x \geq 1$, on $[0, 2]$. Computing piece by piece, $\int_0^2 f = 0 + 1 = 1$, so the average is $\frac{1}{2}$ — a value f never takes.
3. Since both sides of $\int_a^b f = f(c)(b-a)$ change sign when a and b are interchanged (Definition 13.1), the theorem also holds for $a > b$, with c between b and a .

13.8 Summary

- The integral of a continuous function is defined whether or not $f \geq 0$; in general it is a **signed area** (Remark 13.1), and positive and negative contributions can cancel (Example 13.1). The **total area** is $\int_a^b |f|$.
- **Additivity** (Proposition 13.1): $\int_a^b f = \int_a^c f + \int_c^b f$ for $a < c < b$. Requiring that additivity survive forces $\int_a^a f = 0$ and $\int_b^a f = -\int_a^b f$ (Definition 13.1): integrating from right to left counts area negatively.
- **Linearity**: $\int(f + g) = \int f + \int g$ and $\int cf = c \int f$ (Proposition 13.2, Proposition 13.3), both straight from the Riemann sums and the limit laws. There is no such rule for products (Remark 13.2).
- Together with the **integrals of elementary functions** (Proposition 13.4), linearity lets us integrate all linear combinations of them (Example 13.3).
- **Estimates**: $f \geq 0 \Rightarrow \int f \geq 0$ and $f \geq g \Rightarrow \int f \geq \int g$ (Proposition 13.5, Proposition 13.6). For continuous f with minimum m and maximum M , $m(b - a) \leq \int_a^b f \leq M(b - a)$ (Corollary 13.2). Applied to $\frac{1}{x}$ on the intervals $[k, k + 1]$, this shows $1 + \frac{1}{2} + \cdots + \frac{1}{n} \geq \ln(n + 1) \rightarrow \infty$ (Example 13.4).
- $\frac{1}{b-a} \int_a^b f$ is the **average value** of f on $[a, b]$ (Definition 13.2), the limit of the averages of f over n equally spaced points. By the Intermediate Value Theorem it is actually attained: $\int_a^b f = f(c)(b - a)$ for some $c \in [a, b]$ (Theorem 13.1).

14 The Indefinite Integral

Companion reading

This lecture corresponds to C&J §2.4 (the integral as a function of the upper limit).

So far we have treated the integral $\int_a^b f(x) dx$ (Definition 11.2) as a *number* attached to a function and an interval. But this number depends on the limits of integration a and b . If we hold the lower limit fixed and let the upper limit vary, the integral becomes a *function* of that upper limit — and it is worth asking what kind of function we get.

14.1 Definition and Basic Formula

To make clear that the upper limit is now the variable, we rename it x and use a different letter, say u , for the variable of integration. (Recall that the name of the integration variable is irrelevant: $\int_a^b f(x) dx = \int_a^b f(u) du$.)

Definition 14.1.

Let f be continuous on an interval I , and let $\alpha \in I$. The function

$$\Phi_\alpha(x) = \int_\alpha^x f(u) du \quad (x \in I)$$

is called **an indefinite integral** of f . For $x < \alpha$ and $x = \alpha$ the integral is understood in the sense of Definition 13.1; in particular $\Phi_\alpha(\alpha) = 0$.

Remark 14.1. Note the article: *an* indefinite integral, not *the* indefinite integral. If we choose a different lower limit β , we obtain a different indefinite integral,

$$\Phi_\beta(x) = \int_\beta^x f(u) du.$$

As we will see in a moment, the choice of the lower limit is not that important. So we fix α once and for all and simply write

$$\Phi(x) = \Phi_\alpha(x).$$

Proposition 14.1 (Basic properties of the indefinite integral). *Let f be continuous on I and $\Phi(x) = \int_{\alpha}^x f(u) du$.*

1. *If we know Φ , we can compute **any** definite integral of f : for all $a, b \in I$,*

$$\int_a^b f(x) dx = \Phi(b) - \Phi(a).$$

2. *Any other indefinite integral differs from Φ by a constant: for every $\beta \in I$,*

$$\Phi_{\beta}(x) = \int_{\beta}^x f(u) du = \Phi(x) - \Phi(\beta).$$

Proof.

- (1) By additivity of the integral (Proposition 13.1, in the general form of Exercise 13.1, since a, b, α may lie in any order) and by Definition 13.1,

$$\begin{aligned} \int_a^b f(x) dx &= \int_a^{\alpha} f(x) dx + \int_{\alpha}^b f(x) dx \\ &= - \int_{\alpha}^a f(x) dx + \int_{\alpha}^b f(x) dx \\ &= \Phi(b) - \Phi(a). \end{aligned}$$

- (2) This is the special case $a = \beta, b = x$ of (1).

□

Remark 14.2. In (2) the constant $-\Phi(\beta)$ depends on β , but **not** on x . So all the indefinite integrals of f have graphs that are vertical translates of one another; changing the lower limit only shifts the graph up or down. Part (1) generalizes Corollary 13.1: there the lower limit a played the role of α .

14.2 Properties of Φ as a Function

What properties does $\Phi(x)$ have as a function? Geometrically, $\Phi(x)$ is the signed area (Remark 13.1) under the graph of f between α and x , and moving x to the right adds the signed area over the new piece of the interval.

Proposition 14.2.

- If $f \geq 0$ on I , then Φ is increasing (1): as x gets bigger we add positive area.
- If $f \leq 0$ on I , then Φ is decreasing: as x gets bigger we add negative area.

Proof. Let $x < y$ in I (for $x = y$ there is nothing to show). By Proposition 14.1,

$$\Phi(y) - \Phi(x) = \int_x^y f(u) du.$$

If $f \geq 0$, this integral is ≥ 0 by Proposition 13.5, so $\Phi(x) \leq \Phi(y)$. If $f \leq 0$, apply Proposition 13.5 to $-f \geq 0$ and use Proposition 13.3 to get $\int_x^y f(u) du \leq 0$, so $\Phi(x) \geq \Phi(y)$. $\square$

14.2.1 Continuity of the indefinite integral

Theorem 14.1.

If f is continuous on I , then Φ is continuous on I .

Proof. Let $x, y \in I$. By Proposition 14.1 and the Mean Value Theorem for Integrals (Theorem 13.1, which requires f continuous; for $y < x$ use Remark 13.4, part 3),

$$\Phi(y) - \Phi(x) = \int_x^y f(u) du = f(c) (y - x) \quad (*)$$

for some c between x and y .

What happens if $y \rightarrow x$? For Φ to be continuous at x , we need $\Phi(y) \rightarrow \Phi(x)$ (Proposition 10.2). Using (*) and the limit laws (Proposition 10.3),

$$\begin{aligned} \lim_{y \rightarrow x} \Phi(y) &= \lim_{y \rightarrow x} [\Phi(x) + f(c) (y - x)] \\ &= \lim_{y \rightarrow x} \Phi(x) + \lim_{y \rightarrow x} f(c) (y - x) \\ &= \Phi(x) + f(x) \cdot 0 = \Phi(x). \end{aligned}$$

$\square$

Remark 14.3. The point c in (*) depends on y , so a little care is needed in the last step. Since c lies between x and y , we have $c \rightarrow x$ as $y \rightarrow x$, and hence $f(c) \rightarrow f(x)$ by continuity of f . In fact, all we really need is that $f(c)$ stays **bounded** while $y - x \rightarrow 0$ — which is exactly what the next result exploits.

Proposition 14.3 (Lipschitz continuity). *Let f be continuous on I , let $[a, b] \subseteq I$, and let $M = \max_{[a, b]} |f|$ (this maximum exists by the Extreme Value Theorem, see Remark 11.2). Then for all $x, y \in [a, b]$,*

$$|\Phi(y) - \Phi(x)| \leq M |y - x|.$$

*A function satisfying an estimate of this kind is called **Lipschitz continuous**.*

Proof. By (*), $|\Phi(y) - \Phi(x)| = |f(c)| |y - x|$ with c between x and y , so $c \in [a, b]$ and $|f(c)| \leq M$. $\square$

Remark 14.4. Lipschitz continuity is a stronger property than continuity: given $\varepsilon > 0$, the choice $\delta = \varepsilon/(M + 1)$ works at **every** point of $[a, b]$ simultaneously (compare Example 6.3, where the same kind of δ appeared). In other words, Φ is *uniformly continuous* on $[a, b]$ in the sense of Equation 11.1, and here we get this directly, with an explicit δ .